Grafting: Decoupled Scale Factors and Modulus in RNS-CKKS

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Abstract

The CKKS Fully Homomorphic Encryption (FHE) scheme enables approximate arithmetic on encrypted complex numbers for a desired precision. Most implementations use RNS with carefully chosen parameters to balance precision, efficiency, and security. However, a key limitation in RNS-CKKS is the rigid coupling between the scale factor, which determines numerical precision, and the modulus, which ensures security. Since these parameters serve distinct roles—one governing arithmetic correctness and the other defining cryptographic structure—this dependency imposes design constraints, such as a lack of suitable NTT primes and limited precision flexibility, ultimately leading to inefficiencies.

We propose *Grafting*, a novel approach to decouple scale factors from the modulus by introducing *(universal) sprouts*, reusable modulus factors that optimize word-sized packing while allowing flexible rescaling. The universal sprouts allow rescaling by arbitrary bit-lengths and key-switching at any modulus bit-length without requiring additional key-switching keys, and thus enable universal use of the underlying parameters. Decoupling the scale factor from the modulus in Grafting yields significant efficiency gains: (1) Optimized RNS packing by decomposing the modulus into machine word-sized components, accelerating computations and reducing the ciphertext and encryption/evaluation key sizes; and (2) A freely adjustable scale factor independent of the modulus, unifying the ring structure across applications and reducing modulus consumption through adaptive scalings.

Our experiments demonstrate that Grafting improves performance across standard SHE/FHE parameter sets for ring dimensions 2^{14}-2^{16} by up to 1.83× and 2.01× for key-switchings and multiplications, respectively, and up to 1.92× for bootstrapping. Grafting also reduces public key and ciphertext sizes by up to 62% without compression, maintaining the same number of public keys as before. As an application, we showcase the CKKS gate bootstrapping for bits (Bae et al; Eurocrypt'24), achieving 1.89× speed-up due to the reduced number of RNS factors. Finally, we revisit the homomorphic comparison (Cheon et al; Asiacrypt'20), evaluating it with carefully chosen scale factors for each iteration, reporting up to 204-bit fewer modulus consumption (27% reduction) in the standard parameter set, without precision loss.

CCS Concepts

• **Security and privacy → Cryptography**.

Keywords

Homomorphic Encryption, CKKS, Residue Number System, Bootstrapping

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1 Introduction

The Cheon–Kim–Kim–Song (CKKS) scheme [17] is a Fully Homomorphic Encryption (FHE) scheme that allows approximate computation over real/complex-valued encrypted data. For real/complex-valued computations, the precision is determined by the *scale factors* used for each homomorphic operation. It is multiplied with the cleartext messages and rounded, encoding the real/complex messages into integers via Discrete Fourier Transform (DFT). Therefore, the sequence of scale factors becomes a key parameter to ensure arithmetic correctness at the desired precision level.

CKKS bases its IND-CPA security on the hardness of the Ring-Learning-With-Errors (RLWE) problem, where the modulus budget is limited for security. As in other RLWE-based FHE schemes, a huge modulus budget of hundreds to thousands of bits is used for its underlying cryptographic structure, the polynomial ring. To efficiently handle the huge modulus, every competitive library [2, 21, 33, 40] uses the Residue Number System (RNS) and Number Theoretic Transform (NTT) [16], decomposing the huge modulus into RNS factors, each of which is an NTT prime. The use of RNS in CKKS enables significantly more efficient homomorphic operations but introduces some restrictions: the linkage between the scale factors and the RNS factors. As the scale factors are multiplied during homomorphic multiplications, it necessitates *rescaling* to revert to their original size. This procedure not only scales down the scale factors but also the ciphertext modulus, consuming the modulus budget. Thus, an RNS factor should exist in the modulus, which has a size similar to the scale factor; otherwise, it introduces a larger error that sometimes exponentially explodes [1, 16, 29], necessitating a careful selection of the RNS factors.

In RNS-CKKS, the two key parameters–the scale factor, which governs computational accuracy, and the ciphertext modulus, which provides a cryptographic structure–are tightly coupled, significantly restricting flexibility in parameter selection. Ideally, the scale factor should be adjusted according to the application's required precision to minimize modulus consumption. However, since the scale factor is intrinsically linked to the modulus structure, any change requires regenerating not only the ciphertext modulus but also the key modulus and the public keys accordingly, adding substantial management overhead. As a result, most deployments rely on general-purpose or pre-configured parameter sets.

Recent advances, however, have explored application-dependent parameter optimization by fine-tuning scale factors to match precision needs more precisely. In bootstrapping, for instance, performance can be improved by assigning different scale factors to each step based on individual precision requirements, which results in faster execution and more efficient use of the available modulus [21, 33]. This leads to better throughput in terms of amortized bootstrapping time per available level. Also, the use of fine-grained scale factors for different purposes reduces the overall modulus consumption [13, 32, 41], resulting in better running time.[1] It is also demonstrated that highly efficient low-precision operations using small scale factors (e.g., 25–35 bits) can be designed [4, 5], enabling the use of smaller ring dimensions, yielding benefits in both speed and memory footprint.

However, due to its bonded nature, it is unable to construct parameters with arbitrary scale factors for RNS-CKKS. In particular, there are not enough small NTT primes for typical CKKS ring dimensions of 2^{14} to 2^{17}, limiting the parameter designs. Furthermore, when it comes to hardware implementation, we suffer from the smaller word size of the machine, introducing another restriction to the size of the scale factors to be less than the machine's word size, say, 32 bits.

Furthermore, the RNS factors, which align with the scale factor sizes, deteriorate both performance and memory efficiency, hindering the optimal use of the machine's resources. Since CKKS scale factors typically range from 30 to 60 bits, the ciphertext modulus must be decomposed into RNS primes of similar sizes. When these scale factors are utilized, the number of RNS factors significantly increases compared to the optimal case. Given that the performance of homomorphic operations scales linearly (and sometimes quadratically) with the number of RNS factors, this increase leads the overall performance to be far from optimal.

Until now, decoupling the scale factors and the ciphertext modulus has been a challenging problem. Precisely, the CKKS ciphertext is a tuple (a, b) over a polynomial ring $\mathcal{R}_Q = \mathbb{Z}[X]/(X^N+1)$, where N is the ring dimension, which is a power-of-two integer and Q is a positive integer modulus. Each ciphertext has its multiplicative level ℓ, corresponding modulus Q_ℓ, and scale factor Δ_ℓ. Rescaling at level ℓ essentially divides the squared scale factor Δ_ℓ^2 by q_ℓ, resulting in a scale factor $\Delta_{\ell-1} = \Delta_\ell^2/q_\ell$. To slightly detour this relationship, Agrawal et al. [1] introduced the composite rescaling using the composition of the RNS factors instead of a single RNS factor q_ℓ, to make the moderate precision RNS-CKKS available in the hardware implementation. However, this does not actually decouple the scale factors and the modulus; it allows the modulus factors to be crafted to compose them roughly the size of the scale factor. As a side effect, the number of RNS factors increases, leading to a slower running time in the larger word-size machines, e.g., 64-bit CPUs, but a better performance in the machines with smaller word sizes, e.g., 32-bit GPUs.

Another approach of Mono et al. [37] was to adopt the nature of using mostly word-sized primes from the Double-CRT format in BGV [26] to RNS-CKKS. The ciphertext modulus can be switched from one to another having a similar size while ensuring that the new modulus has a similar size to control the size of the error newly introduced by the modulus switching. For example, when the scale factor is 36-bit, one can construct a ciphertext modulus with 54-bit NTT primes, and replace two 54-bit primes with three 36-bit primes before rescaling. This allows for a smaller number of RNS factors in the ciphertext modulus and decreases the overall computation time. Despite that, this approach reduced the number of RNS factors and improved the homomorphic computation efficiency; however, it was restricted to particular cases and pre-determined scale factors.

Samardzic and Sanchez [39] extended this approach, suggesting a similar concept of modulus managing technique, focusing on designing a hardware accelerator. Fusing the modulus switching with the CKKS rescaling,[2] the ciphertext modulus can be rescaled with a rational factor, from Qa to Qb, for some greatest common divisor $Q \in \mathbb{Z}$ of the moduli and the integer factors a and b,

[1] via decreasing the number of levels used, or the number of required bootstrappings.

[2] This concept was concurrently introduced in an earlier version of this work.

where $a/b \approx \Delta$. Given the largest possible modulus and a sequence of desired rescaling factors, one may find a descending chain of ciphertext moduli, where each modulus is a composition of the machine's word-size NTT primes with possibly several smaller NTT primes. Again, the application-dependent scale factors must be pre-determined for public key generation. Also, it is not compatible with the RNS-CKKS key-switching algorithm, the most frequent and heavy operation in HE, which is used for homomorphic multiplication or rotation. For key-switching, the ciphertext moduli need to be divisible by one another so that the least common multiple (LCM) of the ciphertext moduli becomes a factor of the key-switching key modulus. Without divisibility, the LCM becomes too large, pushing the key-switching key modulus beyond the secure budget. To stay within the secure modulus budget, multiple key-switching keys must be generated for each operation type–e.g., several relinearization keys for different multiplicative levels or multiple rotation keys for each rotation amount. This significantly increases the size of the public key, which already ranges from hundreds of megabytes to tens of gigabytes, amplifying the communication overhead in the typical secure outsourced computation scenarios.

None of the prior approaches efficiently resolves the tight coupling between the scale factors and the ciphertext modulus, and the following question arises:

*C*an we completely decouple scale factors from modulus,
increase efficiency of homomorphic computations,
while still preserving key-switching key sizes?

1.1 Our Contribution

We introduce *Grafting*, a modulus management framework for RNS-CKKS that completely decouples scale factors from the ciphertext modulus. This separation enables full utilization of the machine's computational and memory capabilities by allowing an optimal number of RNS factors. As a consequence, Grafting yields both a run-time speed-up and a significant reduction in the size of public keys proportional to the decrease in the number of RNS factors. Furthermore, the full decoupling allows application-independent parameterization–a single parameter set can universally support a wide range of applications. Moreover, unlike conventional schemes where scale factors must be rigidly determined during public key generation, Grafting supports flexible selection of scale factors even after the public parameters and the keys are fixed. This flexibility leads to substantial savings in modulus consumption, further improving overall efficiency.

High-level Overview of Grafting. Grafting is built on top of the RNS-CKKS multiplications and key-switchings, but manages the ciphertext and switching key moduli differently. As a useful tool, we first introduce the rational version of the CKKS rescale operation, rescaling by a rational number instead of an integer, similar to [39], namely the *rational rescale*. It is a composition of integer multiplication, and the usual CKKS rescale, i.e., modulus changing from Qa to Qab, then rescaled down to Qb, at the same time, the scale factor is also updated from Δ to $\Delta \cdot b$, then $\Delta \cdot b/a$ for some integers Q, a, and b. We proved that the error introduced by rational rescaling is the same amount as the usual rescaling, allowing the ciphertext modulus to be decomposed with the machine's word-sized factors as much as possible. However, this is not enough for efficient key-switching, requiring the divisibility between different ciphertext moduli. We introduce *(universal) sprout*, a small and reusable part of the ciphertext modulus, so that it can introduce divisibility to the ciphertext moduli. For instance, we can have a power-of-two integer 2^α as an RNS factor, composing the modulus $Q = q_0 q_1 \cdots q_\ell \cdot 2^\alpha$ with the word-size primes q_i for an integer α smaller than the machine word's bit-length, ω. Any two moduli can divide one another, allowing key-switchings as in RNS-CKKS. Based on this, we revisit the RNS-CKKS key-switching algorithms to allow the RNS factors of small power-of-two integers while keeping the NTT states. We illustrate how the modulus changes in Grafting in Figure 1.

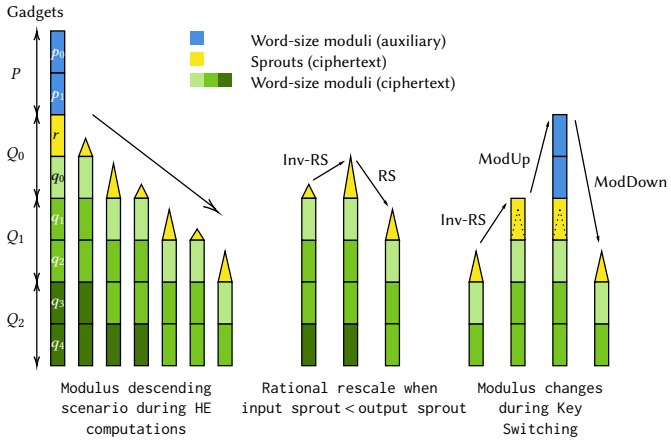


Figure 1: Modulus Changes in Grafting for Ciphertexts. The blue (p_i) and the green (q_j) rectangles are word-size prime factors, and the yellow rectangles (r) and triangles are sprouts. Each modulus is composed of word-size primes and a sprout.

Applications. As Grafting breaks the linkage between the scaling factors and the RNS factors, it can solve many restrictions that previous implementations had. Grafting optimally packs the modulus using mostly machine word-sized factors, resulting in fewer RNS components. This reduction not only accelerates homomorphic operations but also decreases the ciphertext and public key sizes, roughly in proportion to the reduced number of factors. In this respect, we revisit the Somewhat HE (SHE) and Fully HE (FHE) parameters in the literature [2, 8, 21, 33] and redesign their grafted variants with reduced RNS factors by up to 41%.

Also, we do not need to suffer from the lack of small NTT primes or scale factors larger than the machine's word size, which allows optimized performance for lower—and/or higher-precision computation. In addition, Grafting is particularly effective in applications that use smaller-than-usual scale factors [4, 5, 13] where the original number of RNS factors is far from optimal. In this regard, we revisit the Bit-CKKS [4], which uses the scale factors of sizes down to 26 bits. The grafted implementation can reduce the number of RNS factors by up to 46%. Further exploiting the benefits of Grafting, we introduce an additional parameter set in $\log_2 N = 14$,

Table 1: Gains obtained in the standard SHE [8], FHE [21], and Bit-CKKS [4] parameter sets using Grafting.

Parameter	Speed-ups			Key-size
	KeySwitch	Mult	BTS	reduction
SHE15 [8]	1.29×	1.42×	-	42% ↓
FHE15s [21]	1.72×	1.78×	1.92×	62% ↓
BinFHE16 [4]	1.92×	2.07×	1.81×	25% ↓

which cannot be implemented in RNS-CKKS–3 available levels after binary gate bootstrapping instead of 2 available levels.

On the other hand, because of the decoupled nature, the modulus can be independent of the application. In this respect, in Grafting, a single underlying parameter can be used universally in different applications and/or precisions. Further, the scale factors can be freely changed for each sub-procedure of homomorphic computations. Like adaptive-precision techniques in cleartext computation, such as those used in machine learning training/inference or Newton-like iterative methods, the encrypted counterpart also enhances performance by reducing overall modulus consumption through carefully tuned scale factors at each sub-procedure. As an example, we revisit the iteration-based homomorphic comparison technique [18], utilizing fine-grained scale factors for each iteration, ensuring the accuracy of the homomorphic comparison. We theoretically demonstrate that the scale factors can be smaller in the early iterations, reducing the modulus consumption, possibly reducing the number of bootstrappings in wider applications such as the ReLU function evaluation for neural networks [34, 35, 38] or lookup table evaluations [15, 22].

Implementation and Experimental Results. We implemented Grafting with RNS-CKKS in C++, relying on most of the well-known optimization techniques. To compare the performance, we also implemented the non-grafted variant; we will refer to it as a *simple* variant. Based on the redesigned parameters for SHE and FHE, we achieved speed-ups of up to 2.01× for multiplications and 1.92× for bootstrappings, and reduced the key sizes by up to 62% without compression. Note that the available modulus after bootstrapping and the precision of the homomorphic operations remain the same as before.

For Bit-CKKS [4], we achieved speed-ups of 1.89× and 1.81× for ring dimensions 2^{14} and 2^{15}. The additional parameter set reports similar timings with the original $\log_2 N = 14$ parameter set, but with one more available level.

We summarize the main results in Table 1. Please see Section 5 for more details.

For the homomorphic comparison, we analyze the error behaviors during the polynomial evaluations and pick the scale factors that guarantee the precision as high as before. With the smaller scale factors, we achieved 10-27% less modulus consumption compared to the prior art for evaluating two different homomorphic comparison functions without precision loss. We also demonstrate a reduction in the number of bootstrapping operations for homomorphic comparison evaluation, thanks to the reduced modulus consumption, which naturally accelerates execution by a factor of $3/2 = 1.5\times$.

Implications for Other FHE Schemes. While Grafting primarily focuses on the CKKS FHE scheme due to its real-valued computation, its modulus management extends naturally to other RNS-based schemes, including cases where scale factors and moduli are not tightly coupled. For example, BGV also benefits from more precise scaling, which consumes fewer bits, and from using fewer RNS moduli during key switchings. Moreover, Grafting can be applied to accelerate other FHE bootstrappings, as recent works demonstrate that CKKS bootstrapping can be leveraged to accelerate BFV or batched TFHE bootstrapping [4, 5, 30].

Additional Details and Applications. We provide several further contributions[3], including details on Grafting implementation, *grafted* Tuple-CKKS [13], arbitrary-precision computation, and revisited parameters from the literature.

1.2 Additional Related Works

In the introduction, we already mentioned the relevant related works to our work. Following, we present additional related works.

RNS-CKKS. The original CKKS [17] scheme used the power-of-two integers for its ciphertext modulus; however, due to its limited efficiency, the RNS was introduced for CKKS implementations [16], as in other RLWE-based HE schemes like BGV [12, 24] or B/FV [6, 11, 23] schemes. Currently, most of the available homomorphic encryption libraries, indeed, every competitive library, are based on RNS for efficiency [2, 21, 25, 33, 40]. In RNS-CKKS, the ciphertexts and the switching keys are decomposed into integers smaller than the word size; each is a remainder modulo an RNS factor. The computation is done in a decomposed manner, thanks to the CRT. From its RNS-friendly nature, Han and Ki [28] introduced an advanced key-switching technique for RNS-CKKS, adopted from [6, 24], which gave the flexibility of choosing the usable ciphertext modulus, compensating for the size of the switching keys and the key switching running times. Using the so-called RNS gadget decomposition, the ciphertexts are decomposed into several blocks and multiplied by the corresponding switching keys. By elaborately designing the modulus, its RNS factors, and the gadgets, more efficient bootstrappings can be achieved, accelerating the overall homomorphic computations [9, 29, 32] and libraries [21, 33]. However, as mentioned earlier, the relationship forces the RNS factors to be set far smaller than the machine's word size, making it hard to fully utilize the machine's computation and memory budget.

Approaches Filling the Machine Word Sizes. The idea to use word-sized primes mostly and a few small primes in the RNS factors was first proposed by Gentry, Halevi, and Smart [24] from their BGV implementation [26]. They introduce a method of choosing word-size primes and some smaller primes, enabling the ciphertext modulus to be fairly close to any desired target value. For key-switching, they switch the modulus by putting more word-sized primes and dropping non-word-sized primes, maintaining the same message. However, this approach is not applicable to CKKS, as the message is not well-preserved, unlike in BGV. This induces an absolutely larger error during modulus-switching unless the input/output ciphertext moduli are extremely close.

[3] can be found in the full version of this paper [14].

In [31], the machine word-sized moduli were partly utilized for the key-switching operation. However, their technique is advantageous only for sufficiently large gadget ranks and applies only to the key-switching operation. Grafting, however, accelerates the whole homomorphic computations with no such restrictions on gadget ranks.

A following work by Belorgey et al. [7] extended their technique to digit-based gadget decompositions and proposed to implement FHE using binary modulus with Discrete Fourier Transform (DFT) instead of NTT. However, this requires higher precision and increases the number of DFT units, leading to less efficient implementation.

Machine-dependent Approaches. Agrawal et al. [1] proposed an implementation depending on the machine word size to design a 32-bit hardware implementation with the RNS-CKKS parameters. As the scaling factors range from 48 to 58 bits, two NTT primes of 24 to 29 bits were required for each rescale operation. It is worth noting that the smaller primes are hard to use since there are fewer NTT primes, such as 24 to 29 bits, lacking NTT primes for larger ring degrees. This was also the case in Tuple-CKKS [13], where the performance was restricted due to the non-existence of NTT primes of appropriate sizes.

2 Preliminaries

2.1 Notations

Polynomials are denoted in bold font and lowercase letters. We let $\lfloor y \rceil$ be a rounding of $y \in \mathbb{R}$ to the nearest integer. We naturally extend the rounding notation to vectors and polynomials by applying it component-wise. For an integer n, we denote a set of non-negative integers smaller than n as $[n]$, i.e., $[n] = \{0, 1, \cdots, n-1\}$.

We let $\mathcal{R} = \mathbb{Z}[X]/(X^N+1)$ be a polynomial ring where N is a power-of-two integer. For any positive integer Q, let the quotient ring $\mathcal{R}_Q = \mathcal{R}/Q\mathcal{R} = \mathbb{Z}_Q[X]/(X^N+1)$. We let $\mathsf{ct} = (\mathbf{b}, \mathbf{a}) \in \mathcal{R}_Q^2$ be a ciphertext with ciphertext modulus Q with respect to a secret key $\mathbf{s} \in \mathcal{R}$ if it satisfies $\langle \mathsf{ct}, (1, \mathbf{s}) \rangle \approx \mathbf{m} \mod Q$ for some message $\mathbf{m} \in \mathcal{R}$ with respect to a target message precision. For a positive integer q and a polynomial $\mathbf{a} \in \mathcal{R}$ (or $\mathcal{R}_Q$ for some $q \mid Q$), we let $[\mathbf{a}]_q$ be a representative of $\mathbf{a}$ mod q in $\mathcal{R}_q$. For positive integers r and Q, we denote $\mathrm{ord}_r\, Q$ as the largest integer k such that r^k divides Q.

2.2 Number Theoretic Transform

For a polynomial $\mathbf{a} = a_0 + a_1 x + \cdots + a_{N-1}x^{N-1} \in \mathcal{R}_q$, we let $\mathrm{NTT}(\mathbf{a}) = (\mathbf{a}(\zeta^0), \mathbf{a}(\zeta^1), \cdots, \mathbf{a}(\zeta^{N-1})) \in \mathbb{Z}_q^N$ be a number theoretic transform (NTT) of $\mathbf{a}$ in modulus q, where $\zeta \in \mathbb{Z}_q$ is a primitive N-th root of unity in $\mathbb{Z}_q$, which only exists when $2N \mid (q-1)$. We call primes q satisfying the condition $2N \mid (q-1)$ the NTT primes (with respect to ring dimension N and modulo q). For a vector $\mathbf{b} = (b_0, \cdots, b_{N-1}) \in \mathbb{Z}_q^N$, we let $\mathrm{iNTT}(\mathbf{b}) = \sum_{i=0}^{N-1} \tilde{b_i} x^i$ be an inverse NTT (iNTT), where $\tilde{b_i} = n^{-1} \cdot \sum_{j=0}^{N-1} b_j \cdot \zeta^{-ij} \in \mathbb{Z}_q$. Note that NTT and iNTT commute, i.e.,

$$\mathrm{NTT}(\mathrm{iNTT}(\mathbf{b})) = \mathbf{b}, \text{ and } \mathrm{iNTT}(\mathrm{NTT}(\mathbf{a})) = \mathbf{a},$$

and that NTT and iNTT are homomorphic. We let the coefficient vector $(a_0, a_1, \ldots, a_{N-1}) \in \mathbb{Z}_q^N$ be an *NTT-coefficient* format of $\mathbf{a}$, and $\mathrm{NTT}(\mathbf{a}) \in \mathbb{Z}_q^N$ be an *NTT-evaluated* format of $\mathbf{a}$.

2.3 Computation in RNS-CKKS

The RNS-CKKS scheme is an RNS variant of the CKKS scheme, first introduced in [16]. The ciphertext and the switching key modulus comprise NTT primes, constituting the RNS factors. Specifically, $Q_{\max} = q_0 \cdots q_L$ be the maximum ciphertext modulus, where q_i are relatively prime NTT primes. The ciphertext modulus is $Q = q_0 \cdots q_\ell$ for some $\ell \in [L+1]$. The switching key modulus is $PQ_{\max}$, where $P = p_0 \cdots p_{K-1}$, where p_j's are relatively prime NTT primes. In addition, q_i's and p_j's are relatively prime. The polynomials in $\mathcal{R}_Q$ are stored and computed in RNS, i.e., for $a \in \mathcal{R}_Q$, we indeed have $[\mathbf{a}]_{q_i} \in \mathcal{R}_{q_i}$ for every RNS factor $q_i | Q$. We note that the CRT decomposition is homomorphic.

We now recall some main features of the RNS-CKKS scheme.

2.3.1 Fast Basis Conversion in [16]. Let $\mathcal{B} = \{p_0, \ldots, p_{k-1}\}$ and $\mathcal{C} = \{q_0, \ldots, q_{l-1}\}$ be the bases for moduli $P = \prod_{i=0}^{k-1} p_i$ and $Q = \prod_{j=0}^{l-1} q_j$, respectively, where the base moduli are pairwise relatively prime. A RNS representation of an element $\mathbf{a} \in \mathbb{Z}_Q$ is denoted by

$$[\mathbf{a}]_{\mathcal{C}} = (\mathbf{a}^{(0)}, \ldots, \mathbf{a}^{(l-1)}) \in \mathbb{Z}_{q_0} \times \cdots \times \mathbb{Z}_{q_{l-1}} \ .$$

One can convert such $\mathbf{a}$ into its RNS representation with respect to $\mathbb{Z}_P$ as

$$\mathsf{Conv}_{\mathcal{C}\to\mathcal{B}}([\mathbf{a}]_{\mathcal{C}}) = \left(\sum_{j=0}^{\ell-1} [\mathbf{a}^{(j)} \cdot \hat{q}_j^{-1}]_{q_j} \cdot \hat{q}_j \pmod{p_i} \right)_{0 \le j < k},$$

where $\hat{q}_j = Q/q_j$. Note that $\tilde{\mathbf{a}} := \sum_{j=0}^{\ell-1} [\mathbf{a}^{(j)} \cdot \hat{q}_j^{-1}]_{q_j} \cdot \hat{q}_j = \mathbf{a} + Qe$ for some small $e \in \mathbb{Z}$ satisfying $|\tilde{\mathbf{a}}| \le (\ell/2) \cdot Q$. Let us use the notation $Q \to P$ instead of $\mathcal{C} \to \mathcal{B}$ if there is no confusion.

2.3.2 Modulus Switching. For a polynomial $\mathbf{a} \in \mathcal{R}_Q^2$, we define the ModUp procedure so that the resulting polynomial is in $\mathcal{R}_{PQ}$, but with the same value in modulus Q and not too large. ModDown reduces the modulus from PQ to Q. It reduces the size of the polynomial and the modulus with the same factor, i.e., by a factor of $Q/PQ \sim P^{-1}$. RS is the same as ModDown but is rescaled by fewer moduli factors than ModDown. It reduces the size of the polynomial and the modulus by q_ℓ. Let $\mathcal{D} = \mathcal{B} \cup \mathcal{C}$. Precisely,

$$\mathsf{ModUp}_{\mathcal{C}\to\mathcal{D}}(\cdot) : \prod_{j=0}^{\ell-1} \mathcal{R}_{q_j} \to \prod_{i=0}^{k-1} \mathcal{R}_{p_i} \times \prod_{j=0}^{\ell-1} \mathcal{R}_{q_j}$$
$$[\mathbf{a}]_{\mathcal{C}} \to (\mathsf{Conv}_{\mathcal{C}\to\mathcal{B}}([\mathbf{a}]_{\mathcal{C}}), [\mathbf{a}]_{\mathcal{C}}),$$
$$\mathsf{ModDown}_{\mathcal{D}\to\mathcal{C}}(\cdot) : \prod_{i=0}^{k-1} \mathcal{R}_{p_i} \times \prod_{j=0}^{\ell-1} \mathcal{R}_{q_j} \to \prod_{j=0}^{\ell-1} \mathcal{R}_{q_j}$$
$$([\mathbf{a}]_{\mathcal{B}}, [\mathbf{b}]_{\mathcal{C}}) \to [P^{-1}]_{\mathcal{C}} \cdot ([\mathbf{b}]_{\mathcal{C}} - \mathsf{Conv}_{\mathcal{B}\to\mathcal{C}}([\mathbf{a}]_{\mathcal{B}})),$$

and $\mathsf{RS}_{q_{\ell-1}}(\cdot) = \mathsf{ModDown}_{\mathcal{C}\to\mathcal{C}'}(\cdot)$, where $\mathcal{C}' = \mathcal{C} \setminus \{q_{\ell-1}\}$.

We note that the ModUp operation maps $\mathbf{a} \in \mathcal{R}_Q$ to $\mathbf{a}+Q\mathbf{e} \in \mathcal{R}_{PQ}$, where $|\mathbf{e}| \le \ell/2$ from the fast basis conversion. The ModDown operation maps $\mathbf{a} = ([\mathbf{a}]_P, [\mathbf{a}]_Q) \in \mathcal{R}_{PQ}$ to $\mathbf{a}' = P^{-1} \cdot (\mathbf{a} - \tilde{\mathbf{a}}) \in \mathcal{R}_Q$,

where $\tilde{\mathbf{a}} \equiv \mathbf{a} \pmod{P}$ and $\|\tilde{\mathbf{a}}\|_\infty \le (k/2) \cdot P$, resulting $\|\mathbf{a}' - P^{-1} \cdot \mathbf{a}\|_\infty = P^{-1} \cdot \|\tilde{\mathbf{a}}\|_\infty \le k/2$. RS introduces an error of size $\le 1/2$.

When applied to ciphertext, the error becomes multiplied by the secret key $\mathbf{s}$ and thus has an infinity norm of $\le k/2 \cdot (\|\mathbf{s}\|_1 + 1)$ and $\le 1/2 \cdot (\|\mathbf{s}\|_1 + 1)$, respectively, for ModDown and RS. Let us use the notation $Q \to PQ$ (Resp. $PQ \to Q$) instead of $C \to \mathcal{D}$ (Resp. $\mathcal{D} \to C$) if there is no confusion. We note that in [1], it is demonstrated that rather than processing the ModDown at once, splitting the procedure into two or more steps–such as from PQ to $p_0 Q$, and then Q–can reduce the output error by a factor of k with only small additional costs for Hadamard multiplications.

We additionally define the Inv-RS operation, which is sometimes called zero-padding. It multiplies a factor to both the ciphertext and its modulus and is used during key switching with gadget decomposition.

2.3.3 Inverse Rescale. For given a polynomial $\mathbf{a} \in \mathcal{R}_Q$, we define inverse rescaling by an integer factor R as

$$\text{Inv-RS}_R(\mathbf{a}) = R \cdot \mathbf{a} \in \mathcal{R}_{QR},$$

or in the RNS representation, one can write as:

$$\text{Inv-RS}_R(\mathbf{a}) \equiv \begin{cases} [R]_{q_i} \cdot [\mathbf{a}]_{q_i} & (\bmod\ q_i) \\ 0 & (\bmod\ r_j) \end{cases},$$

for $i \in [\ell]$, $j \in [k]$, where $Q = \prod_{i=0}^{\ell} q_i$ and $R = \prod_{j=0}^{k} r_j$ with co-prime NTT primes q_i's and r_j's. We note that it can be naturally extended to a vector of polynomials or ciphertexts.

2.3.4 Gadget Decomposition and Key Switching. When the ring dimension N, the Hamming weight of the secret key, and the target security are chosen, the maximum possible modulus $PQ_{\max}$ can be determined based on the estimated attack costs of the known attacks via the lattice estimator [3].

The switching key is generated in the largest modulus $PQ_{\max}$, possibly using gadget decomposition. For RNS gadget decomposition, for instance, each gadget Q_i is composed of some RNS factors, and the largest ciphertext modulus $Q_{\max}$ is composed of

$$\begin{aligned} Q_{\max} &= Q_0 \cdots Q_{\mathsf{dnum}-1} \\ &= (q_0 \cdots q_{\alpha-1}) \cdot (q_\alpha \cdots q_{2\alpha-1}) \cdots (q_{\alpha(\mathsf{dnum}-1)} \cdots q_{\mathsf{dnum}\cdot\alpha-1}), \end{aligned}$$

where $\alpha = \lceil (L+1)/\mathsf{dnum} \rceil$ for a pre-fixed parameter $\mathsf{dnum} \in \mathbb{N}$. Note that the relinearization key rlk is a special switching key that switches the secret key from $\mathbf{s}$ to $\mathbf{s}' = \mathbf{s}^2$. We call dnum the gadget rank. We choose P to satisfy $P \gtrsim Q_i$ for all i, so the maximum possible modulus $PQ_{\max}$ should be split into P and Q with roughly $P \approx Q^{1/\mathsf{dnum}}$.

Let $P = p_0 \cdots p_{K-1}$. Then, the switching keys are defined as

$$\{\mathsf{swk}_i\}_{i \in [\mathsf{dnum}]} = \{(\boldsymbol{\beta}_i, \boldsymbol{\alpha}_i)\}_{i \in [\mathsf{dnum}]} \in \mathcal{R}_{PQ}^{2 \times \mathsf{dnum}}, \text{ where}$$
$$\boldsymbol{\beta}_i = -\boldsymbol{\alpha}_i \cdot \mathbf{s} + P \cdot \hat{Q}_i \cdot [\hat{Q}_i^{-1}]_{Q_i} \cdot \mathbf{s}' + \mathbf{e}_i \in \mathcal{R}_{PQ},$$

for $\hat{Q}_i = Q_{\max}/Q_i$, and $\mathbf{e}_i \leftarrow \chi$ be errors. We note that a larger dnum results in a larger usable ciphertext modulus $Q_{\max}$ and a slower key switching operation, also with a larger switching key size.

Key switching. Key switching of a ciphertext $\mathrm{ct} \in \mathcal{R}_Q^2$, denoted as $\mathsf{KeySwitch}_{\mathsf{swk}}(\mathrm{ct})$, involves the following procedures.[4]

(1) Regard the ciphertext in modulus $Q_0 \cdots Q_{d-1}$ instead of $Q = q_0 \cdots q_\ell$, for some α and d satisfying $\alpha(d-1) \le \ell < \alpha d$.
(2) ModUp each component of the ciphertext in modulus Q_i, to $PQ_0 \cdots Q_{d-1}$ for $i \in [d]$, resulting in d ciphertexts in modulus $PQ_0 \cdots Q_{d-1}$.
(3) External product part of each ciphertext with swk_i for $i \in [d]$, and add the remaining parts of the ciphertext.
(4) ModDown from modulus $PQ_0 \cdots Q_{d-1}$ to Q.

2.3.5 Homomorphic Multiplication. Homomorphic multiplication consists of the tensor product of two ciphertexts in the same modulus $Q = q_0 \cdots q_\ell$, then key switch s^2 to s via the rlk, then RS by q_ℓ. Note, the tensor of ciphertexts ct_1 and ct_2 satisfies $\langle \mathrm{ct}_1 \otimes \mathrm{ct}_2, (1, s, s^2) \rangle = \langle \mathrm{ct}_1, (1, s) \rangle \cdot \langle \mathrm{ct}_2, (1, s) \rangle$, where $\mathsf{sk} = (1, s)$.

2.3.6 Level Adjustments. In the RNS setting, we rescale the ciphertexts during multiplication by one of the RNS factors, say Q_i, instead of the real scaling factor Δ. This yields an additional error after a series of rescaling. In [16], it is suggested that choosing each modulus q_i as close as possible to Δ to minimize the error, and later in [29] the authors suggest using level-specific scaling factors. Specifically, the scaling factor Δ_ℓ for each level ℓ is defined as $\Delta_{\ell-1} := \Delta_\ell^2/q_\ell$, iteratively from $\ell = L$ to 1. With different scaling factors in different levels, one may need to manipulate two input ciphertexts having different levels and, thus, different scaling factors. To adjust the ciphertext, the level adjusting technique is introduced [29].

Let ct and ct′ be the ciphertexts with level ℓ and ℓ' ($\ell > \ell'$) and scaling factors Δ_ℓ and $\Delta_{\ell'}$, respectively. Before performing homomorphic operations over ct and ct′, we adjust ct to level ℓ' with the scaling factor $\Delta_{\ell'}$, by Adjust operation: For inputs ct in level $\ell > \ell'$, and the target level ℓ',

(1) Let $\mathrm{ct} = [\mathrm{ct}]_{q_0 \cdots q_{\ell'+1}} \in \mathcal{R}^2_{q_0 \cdots q_{\ell'+1}}$ by dropping the RNS factors $\{q_{\ell'+1}, \ldots, q_\ell\}$,
(2) Multiply a constant $\left\lceil \frac{\Delta_{\ell'} \cdot q_{\ell'+1}}{\Delta_\ell} \right\rfloor$ in $\mathcal{R}_{q_0 \cdots q_{\ell'+1}}$.
(3) RS by $q_{\ell'+1}$.

The resulting ciphertext is in $\mathcal{R}^2_{q_{\ell'}}$ and has a scaling factor $\Delta_{\ell'}$ with an additional error, which is approximately a rounding error.

2.4 CKKS Bootstrapping

As CKKS multiplication consumes the modulus by an amount of the scale factor sizes, CKKS bootstrapping is used to lengthen the multiplicative budget for further computations when the ciphertext modulus reaches the minimum. During bootstrapping, the ciphertext is first homomorphically decoded, then its modulus is raised, and the ciphertext is homomorphically encoded. Then, a reduction modulo the base modulus is applied homomorphically. Each sub-procedures are referred to as SlotToCoeff, ModRaise, CoeffToSlot, and EvalMod.

[4] It requires $(d+2)(\ell+K+1)$ (i)NTT operations, and in particular, $(d+3+2/d)(L+1)$ (i)NTTs at the top level $\ell = L$ [37].

Basically, given an input ciphertext in $\mathcal{R}_{q_0}^2$ encrypting a real-valued message polynomial $\mathbf{m} \in \mathcal{R}$, a series of operations SlotToCoeff, ModRaise, and CoeffToSlot generate a ciphertext in $\mathcal{R}_{Q_{\max}}^2$ encrypting a message polynomial $\mathbf{m} + q_0\mathbf{I}$ for some integer polynomial $\mathbf{I}$. Hence, it requires the moduluo-q_0 operation on the message. Then, during EvalMod, a polynomial approximating the function $x \mapsto (x \mod q_0)$ is homomorphically evaluated. It is approximated as a sine function near the zeros, which is sometimes decomposed into several functions, namely the cosine, double-angle formula, and arcsine functions with proper approximations [10].

3 Grafting: Filling-up machine words in RNS

All the computations in RNS-CKKS are done in the RNS format, i.e., every ciphertext or key is decomposed with respect to RNS factors, and each component is computed with the machine's word size. Therefore, if we can reduce the number of RNS components representing the ciphertext, the whole FHE computation will benefit from a straightforward speed-up by roughly the reduced ratio, depending on the ciphertext modulus.

An appealing approach is to use the word-size primes for the RNS factors. However, it compromises the available multiplicative depths, compared to the cases when the moduli are set approximately the same as the scaling factors, say Δ, which varies (from 20 to 120 in general) on the target message precision.

In this section, we introduce *Grafting*, a method of using word-sized primes for RNS factors while allowing rescaling by an independent factor. We first introduce a tool for switching the ciphertext modulus to a non-divisor modulus, namely, *Rational Rescale* in Section 3.1. We introduce a method to maintain the ciphertext modulus using word-sized NTT primes and a *universal sprout*, enabling optimal performance through *modulus resurrection*. This design supports rescaling to any scaling factor of arbitrary bit-length at any modulus, without incurring additional key generation costs compared to the original RNS-CKKS scheme. As detailed in Section 3.2, this leads to a full decoupling between scale factors and moduli, fully realizing what was only partially addressed in prior works [1, 37, 39]. To support modulus switching for the universal sprout—central to achieving full decoupling—we extend the existing techniques to handle power-of-two factors in the RNS factor, as detailed in Section 3.3.

3.1 Rational Rescale: Rescale with non-divisor

In this section, we decouple the RNS factors from the scaling factors and propose a method to rescale a ciphertext from a modulus Q to another modulus Q', where Q' does not necessarily divide Q, which we call the *Rational Rescaling*. The rational rescale procedure rescales an input ciphertext in modulus Q by a rational number $Q/Q' \in \mathbb{Q}$ to an output ciphertext modulo $Q/(Q/Q') = Q'$, rescaling the scaling factor as well. We note that the same notion was introduced in [39], but without a formal definition or correctness proof. Concretely, we define the rational rescale operation and its correctness as follows.

Definition 3.1 (Rational Rescale). For given a polynomial $\mathbf{a} \in \mathcal{R}_Q$ and $Q' \nmid Q$, we define rational rescaling as the rescaling by a rational factor Q/Q', which can be computed as

$$\mathsf{RS}_{Q/Q'}(\mathbf{a}) = \mathsf{RS}_S(\mathsf{Inv\text{-}RS}_R(\mathbf{a}) \in \mathcal{R}_{\mathrm{lcm}(Q,Q')}) \in \mathcal{R}_{Q'},$$

where $R = \mathrm{lcm}(Q, Q')/Q \in \mathbb{Z}$ and $S = \mathrm{lcm}(Q, Q')/Q' \in \mathbb{Z}$.

Here, the operation RS_S in Definition 3.1 corresponds to the $\mathsf{ModDown}_{\mathrm{lcm}(Q,Q')\to Q'}$ in the original scheme, which rescales by multiple prime factors. In our grafted RNS-CKKS framework, we abuse the notation RS to encompass both the conventional *integer* rescaling—either by a single factor (i.e., the original RS) or multiple factors (i.e., ModDown)—and our extended notion of *rational* rescaling by arbitrary rational factors.

The following theorem shows that the error introduced by rational rescaling is of the same nature as the errors from ModDown and RS operations, i.e., it grows linearly with the number of eliminated RNS factors.

Theorem 3.2 (Rational Rescale Correctness). *For given a ciphertext* $\mathsf{ct} \in \mathcal{R}_Q^2$ *and* $Q' \nmid Q$, *it holds that* $\left[\langle \mathsf{RS}_{Q/Q'}(\mathsf{ct}), \mathsf{sk}\rangle\right]_{Q'} = (Q'/Q) \cdot [\langle \mathsf{ct}, \mathsf{sk}\rangle]_Q + e_{res}$ *for some rescale error* e_{res} *satisfying* $\|e_{res}\|_\infty \le \ell/2 \cdot (\|s\|_1 + 1)$, *where* ℓ *is the number of RNS blocks in* $\mathrm{lcm}(Q, Q')/Q'$, *and* s *be a secret key.*

Proof. Let us follow the notations in Definition 3.1 and let $[\langle \mathsf{ct}, \mathsf{sk}\rangle]_Q = \Delta \cdot m + e$, or $\langle \mathsf{ct}, \mathsf{sk}\rangle = \Delta \cdot m + e + QI$ for some $I \in \mathcal{R}$. Then $\langle \mathsf{Inv\text{-}RS}_R(\mathsf{ct}), \mathsf{sk}\rangle = R\Delta \cdot m + Re + QRI$. The final rescaling RS_S introduces an additional error e_{res}, where $\|e_{\mathrm{res}}\|_\infty \le \ell/2 \cdot (\|s\|_1 + 1)$ for ℓ the number of RNS blocks in $S = \mathrm{lcm}(Q, Q')/Q'$. Precisely, we have

$$\langle \mathsf{RS}_{Q/Q'}(\mathsf{ct}), \mathsf{sk}\rangle = (Q'/Q) \cdot (\Delta \cdot m + e) + Q'I + e_{\mathrm{res}},$$

which concludes the proof. □

Rational rescale maps a ciphertext including a message with a scaling factor Δ^2 (after tensor) to a ciphertext whose message has a scaling factor $\Delta^2 \cdot Q'/Q \approx \Delta$, instead of $\Delta^2/q_\ell \approx \Delta$. We describe homomorphic multiplication with rational rescaling in Algorithm 1. We also note that the rational scaling factors can be tracked, as in [29], holding them as auxiliary data along with the ciphertexts.

Algorithm 1: Homomorphic multiplication.

Input: $\mathsf{ct}_i = (\mathbf{b}_i, \mathbf{a}_i; \Delta) \in \mathcal{R}_Q^2$ for $i = 0, 1$ for the modulus $Q = q_0 \cdots q_{\ell-1} \cdot r$, the scaling factor Δ, and the relinearization key $\mathsf{rlk} \in \mathcal{R}_{PQ_{\max}}^{2\cdot \mathsf{dnum}}$.

Output: $\mathsf{ct} = (\mathbf{b}, \mathbf{a}; \Delta') \in \mathcal{R}_{Q'}^2$, where $Q' = q_0 \cdots q_{\ell'-1} \cdot r'$ and $\Delta' = \Delta^2/(Q/Q')$.

1 $(\mathbf{b}, \mathbf{a}, \mathbf{d}) \leftarrow \mathsf{ct}_1 \otimes \mathsf{ct}_2$ ▷ $\mathbf{b} + \mathbf{as} + \mathbf{ds}^2 \approx \Delta^2 \mathbf{m}_1\mathbf{m}_2$

2 $(\mathbf{b}', \mathbf{a}') \leftarrow \mathsf{KeySwitch}_{\mathsf{rlk}}((0, \mathbf{d}))$ ▷ $\mathbf{b}' + \mathbf{a}'\mathbf{s} \approx \mathbf{ds}^2$

3 $\mathsf{ct} \leftarrow (\mathbf{b}, \mathbf{a}) + (\mathbf{b}', \mathbf{a}')$

4 $\mathsf{ct} \leftarrow \mathsf{RS}_{Q/Q'}(\mathsf{ct})$ ▷ Rational rescale by $Q/Q' \approx \Delta$

return ct

We also propose an algorithm that adjusts two ciphertexts with different moduli and scaling factors before adding or multiplying them, as shown in Algorithm 2. We note that our modulus adjustment is a counterpart of the level adjustment in [29], while

Algorithm 2: Modulus adjustment.

Input: $\mathrm{ct} \in \mathcal{R}_Q^2$ with scaling factor Δ.

Output: $\mathrm{ct}' \in \mathcal{R}_{Q'}^2$ with scaling factor Δ', where $Q > Q'\Delta$.

1 Choose the smallest $Q_{\mathrm{mid}} \mid Q$ such that $Q \geq Q_{\mathrm{mid}} \gtrsim Q' \cdot \Delta$.
2 $\mathrm{ct} \leftarrow (\mathrm{ct} \mod Q_{\mathrm{mid}})$ ▷ Modulo reduction to Q_{mid}
3 $\mathrm{ct} \leftarrow \mathrm{ct} \cdot \lceil (Q_{\mathrm{mid}} \cdot \Delta')/(Q' \cdot \Delta) \rfloor$ ▷ Integer multiplication
4 $\mathrm{ct}' = \mathrm{RS}_{Q_{\mathrm{mid}}/Q'}(\mathrm{ct})$ ▷ Rational rescale to Q'
return ct'

introducing more flexibility by allowing one ciphertext to be adjusted to both the modulus and scaling factor of another.

Lemma 3.3 (Correctness of Modulus Adjustment). *For given an input ciphertext* $\mathrm{ct} = (\mathbf{b}, \mathbf{a}) \in \mathcal{R}_Q^2$ *satisfying* $\mathbf{b} + \mathbf{a} \cdot \mathbf{s} = \Delta \cdot \mathbf{m} + \mathbf{e} \pmod{Q}$*, let moduli* Q' *and* Q_{mid} *satisfy* $Q \geq Q_{mid} \geq Q'\Delta$ *and* $Q_{mid} | Q$*. Then Algorithm 2 outputs a ciphertext* ct' *satisfying* $[\langle \mathrm{ct}', \mathrm{sk} \rangle]_{Q'} = \Delta' \cdot \mathbf{m} + e_{adj}$ *where*

$$\mathbf{e}_{adj} = \frac{\Delta'\mathbf{e}}{\Delta} + \frac{Q'\delta\Delta\mathbf{m}}{Q_{mid}} + \frac{\delta Q'\mathbf{e}}{Q_{mid}} + \mathbf{e}_{res},$$

for some $\delta \in (-1/2, 1/2]$ *and rescale error* $\mathbf{e}_{res}$.

Proof. The modulo reduction does not change the error term, and the integer multiplication only scales up both the messages and the error terms. Let the integer constant be represented as

$$\left\lceil \frac{Q_{\mathrm{mid}}\Delta'}{Q'\Delta} \right\rfloor = \frac{Q_{\mathrm{mid}}\Delta'}{Q'\Delta} + \delta,$$

for some $\delta \in (-1/2, 1/2]$. For the output ciphertext $\mathrm{ct}' = (\mathbf{b}', \mathbf{a}') \in \mathcal{R}_{Q'}^2$ after rational rescaling, it holds that

$$\begin{aligned}
\mathbf{b}' + \mathbf{a}' \cdot \mathbf{s} &= \frac{Q'}{Q_{\mathrm{mid}}} \cdot \left(\frac{Q_{\mathrm{mid}}\Delta'}{Q'\Delta} + \delta \right) \cdot (\Delta\mathbf{m} + \mathbf{e}) + \mathbf{e}_{\mathrm{res}} \pmod{Q'} \\
&= \left(\frac{\Delta'}{\Delta} + \frac{Q'\delta}{Q_{\mathrm{mid}}} \right) \cdot (\Delta\mathbf{m} + \mathbf{e}) + \mathbf{e}_{\mathrm{res}} \pmod{Q'} \\
&= \Delta'\mathbf{m} + \mathbf{e}_{\mathrm{adj}} \pmod{Q'},
\end{aligned}$$

where the $\mathbf{e}_{\mathrm{res}}$ is an error added during rescaling, and $\mathbf{e}_{\mathrm{adj}}$ is as in the theorem. □

We note that the rescale error becomes the most significant one in the adjustment error. We note that the condition $Q_{\mathrm{mid}} > Q' \cdot \Delta$ allows us to manage the adjustment error $\mathbf{e}_{\mathrm{adj}}$ to be sufficiently small; the larger Q_{mid} the lower $\mathbf{e}_{\mathrm{adj}}$.

Once we utilize rational rescaling to manage the scaling factor, however, the resulting ciphertext modulus Q' may not be a divisor of the switching key modulus $PQ_{\max}$. In this case, the original key-switching procedure does not apply to the ciphertext due to the changed modulus, which may not be compatible with the gadget decomposition.

For instance, highlighting the difficulty regarding key-switching, [39] introduces level-specific moduli, called *terminal residues*, which capture the possible bit size of the ciphertext modulus after rational rescaling for each ciphertext level. One might consider preparing keys for all possible ciphertext moduli to support key switching in this setting. However, this will increase the number of switching keys, which degrades the communication cost for key transmissions. When no specific circuits are pre-determined, a huge number of keys need to be transmitted for general circuit evaluations. Having multiple copies is too much because the switching keys are already hundreds to thousands of megabytes for FHE parameters.[5]

3.2 Modulus Resurrection with Universal Sprouts

We propose the *Modulus Resurrection* technique, which enables the use of Rational Rescale without generating additional switching keys or incurring extra costs for key switching, by reusing factors of the ciphertext modulus. As a primary condition for the RNS factors is being relatively prime, we *resurrect* some factors of the top ciphertext modulus, which were scaled out previously. By doing so, the RNS factors remain relatively prime, while the ciphertext modulus is kept to be a divisor of the switching key modulus.

The special resurrecting part of the ciphertext modulus is called *sprout*, which is flexible and possibly small. For all possible sprouts, we define a common multiple r_{top} of the sprouts, which we call the *top sprout*. In other words, we set a maximal sprout r_{top} and choose sprouts from its divisors. Each ciphertext modulus is a product of sprout and distinct word-sized NTT primes, which we call *unit moduli*. In this case, we call the ciphertext modulus *grafted with the sprout*.

The maximum ciphertext modulus is $Q_{\max} = q_0 \cdots q_{L-1} \cdot r_{\mathrm{top}}$ where q_i's be the unit moduli approximately of the machine word size $w = 2^\omega$. Each ciphertext modulus is $Q = q_0 \cdots q_{\ell-1} \cdot r$ for some sprout r. The switching key modulus is $PQ_{\max}$, where P is set conventionally, i.e., $P = p_0 \cdots p_{K-1}$ for relatively prime unit moduli p_i's with $\gcd(P, Q_{\max}) = 1$. As each sprout r is a divisor of r_{top}, it satisfies $r \mid r_{\mathrm{top}}$ and thus $Q \mid Q_{\max} \mid PQ_{\max}$. Hence, we can key switch using the switching keys in modulus $PQ_{\max}$, as many as in the conventional key-switching.

We describe how to construct the sprout r_{top} in an optimal way, in the sense that for any ciphertext modulus Q with an arbitrary integer bit-length, one can always find $\ell \in [L]$ and $r \mid r_{\mathrm{top}}$ such that $Q \approx q_0 \cdots q_{\ell-1} \cdot r$. Then, for any modulus $Q \mid Q_{\max}$ and scaling factor Δ of arbitrary bit-length, we can find $Q' \mid Q_{\max}$ such that $Q' \approx Q/\Delta$. We refer to this property as *universality* and call such a sprout a *universal sprout*, formally defined as follows:

Theorem 3.4 (Universal Rescalability). *Let the ciphertext modulus* $q_i \in [2^\omega(1-\eta), 2^\omega(1+\eta)]$ *for some* $\eta > 0$ *for* $i \in [L-1]$*. Let the maximum sprout modulus* $r_{top} = r_0 \cdots r_s$*. Assume for any positive integer* $\gamma \leq w$*, there exist* $r | r_{top}$ *such that* $r \in 2^\gamma \cdot [1-\epsilon, 1+\epsilon]$ *for some* $\epsilon > 0$*. Then, for any ciphertext in any possible ciphertext modulus* Q*, one can (rational) rescale by* $2^\delta \cdot \left(1 \pm (n\eta + 2\epsilon) + O\left(\eta^2 + \epsilon^2\right)\right)$ *for any positive integer* $\delta < \log_2 Q$*, where* $n = \lceil \delta/w \rceil$.

Proof. Let ct be a ciphertext with modulus $Q = q_0 \cdots q_\ell \cdot r$, where r is a sprout satisfying $r | r_{\mathrm{top}}$, and $r \in 2^\gamma \cdot [1-\epsilon, 1+\epsilon]$, where $0 \leq \gamma < w$. Let $w\ell + \gamma - \delta = w\ell' + \gamma'$, where $0 \leq \gamma' < w$. Since $\delta = w(\ell - \ell') + \gamma - \gamma'$, it holds that $n = \lceil \delta/w \rceil = \lceil \ell - \ell' + (\gamma - \gamma')/w \rceil \geq \ell - \ell'$. Moreover, there exists $r' | r_{\mathrm{top}}$ such that $r' \in 2^{\gamma'} \cdot [1-\epsilon, 1+\epsilon]$ from the assumption. Therefore, for a modulus $Q' = q_0 \cdots q_{\ell'} \cdot r'$,

[5] See Table 5 for practical examples.

we have

$$\frac{1-\epsilon}{1+\epsilon}\cdot(1-\eta)^{\ell-\ell'} \le \frac{Q/Q'}{2^{\delta}} = \frac{q_{\ell'+1}}{2^{w}}\cdots\frac{q_{\ell}}{2^{w}}\cdot\frac{r}{2^{\gamma}}\cdot\frac{2^{\gamma'}}{r'} \le \frac{1+\epsilon}{1-\epsilon}\cdot(1+\eta)^{\ell-\ell'},$$

which concludes the proof. □

Universal Spout Examples. A simple example of a universal sprout is $r_{\text{top}} = 2^{\omega}$, as illustrated below:

Example 3.1 (sprout-60). *Let q_i's be the 60-bit [6] NTT primes and $r_{top} = 2^{60}$. Each sprout r is a power of two integers dividing $r_{top} = 2^{60}$. Thus, any ciphertext modulus of approximately an integer bit can be represented:*

$$Q = q_0 \cdots q_{\ell-1} \cdot 2^{\gamma} \approx (60\cdot\ell+\gamma)\text{-bit modulus},$$

where the top modulus is $Q_{max} = q_0 \cdots q_{L-1} \cdot 2^{60}$. From a ciphertext modulus of approximately $(60\cdot\ell+\gamma)$ bits, one can rational rescale by $q_{\ell'} \cdots q_{\ell-1} \cdot 2^{(\gamma-\gamma')}$ to obtain a ciphertext modulus of approximately $(60\cdot\ell'+\gamma')$ bits, regardless of whether $\gamma \ge \gamma'$.

For instance, if we want to rescale by $\approx$ 36 bits from a modulus $Q = q_0 \cdots q_{\ell-1} \cdot 2^{35}$, we can rational rescale by $(q_{\ell-1}/2^{24}) \approx 2^{36}$, resulting in a ciphertext modulus of $Q' = q_0 \cdots q_{\ell-2} \cdot 2^{59}$.

As in the above example, the top sprout r_{top} is universal if and only if its divisors can approximately represent all the bit lengths from 1 to ω. A universal sprout enables rescaling by nearly any bit-length, allowing it to be used universally, regardless of the target circuit.

We note that efficiently handling a ciphertext modulo 2^{60}, however, is difficult on a 64-bit machine. It requires embedding $\mathcal{R}_{2^{60}}$ into a larger modulus ring $\mathcal{R}_B$, where B must satisfy $B > N \cdot 2^{120}$ to prevent reductions modulo B during multiplication.[7] In practice, such a B involves at least three word-sized moduli, making the computation roughly three times more expensive than using a single 60-bit NTT prime. Instead, we propose another universal sprout in Example 3.2, whose handling cost is roughly twice that of a word-sized NTT prime.

Example 3.2 (sprout-15-16-30). *Let q_i's be 61-bit NTT primes and $r_{top} = 2^{15}\cdot r_1 \cdot r_2$, where r_1 is a 16-bit NTT prime, and r_2 is a 30-bit NTT prime. Typically, we can choose $r_1 = 2^{16}+1$ and $r_2 = 2^{30}-2^{18}+1$, which are NTT primes for ring dimension $N \le 2^{15}$.[8] Each sprout r can be represented as $r = 2^{\alpha}\cdot r_1^{\beta_1}\cdot r_2^{\beta_2}$, where $0 \le \alpha \le 15$, $\beta_i \in \{0,1\}$. We note that the sprouts can represent any bit length from 1 to 61 as*

$$\{2^1, \cdots, 2^{15}, r_1, r_1\cdot 2^1, \cdots, r_1\cdot 2^{13}, r_2, r_2\cdot 2^1, \cdots, r_2\cdot r_1\cdot 2^{15}\}.$$ [9]

It forms a universal sprout that supports rational rescaling with arbitrary bit-length as it satisfies the assumption of Theorem 3.4 with $\epsilon < 2^{-13}$.

For instance, from a modulus $Q = q_0 \cdots q_{\ell-1} \cdot r$, where $r = 2^{13}\cdot r_2 \approx 2^{43}$, let's assume we want to rescale by $\approx$ 15 bits. We can rational rescale by $(r_2/2)$, resulting in a ciphertext modulus $Q' = q_0 \cdots q_{\ell-1} \cdot r'$, where $r' = (2^{13}\cdot r_2)/(r_2/2) \approx 2^{14}$. If we want to rescale by 34 bits in addition, we can rational rescale by $(2^3 \cdot q_{\ell-1}/r_2)$, resulting in a ciphertext modulus $Q'' = q_0 \cdots q_{\ell-2} \cdot r''$, where $r'' = (q_\ell \cdot 2^{14})/(2^3 \cdot q_\ell/r_2) = 2^{11}\cdot r_2 \approx 2^{41}$. Note that a larger amount of rational rescale can be done as well.

In Example 3.2, arithmetic over both $\mathcal{R}_{2^{15}}$ and $\mathcal{R}_{r_1 r_2}$ can be efficiently performed within a single word-sized modulus. In the latter case, this is achieved using composite NTT [20].

Lastly, we propose a third candidate for a universal sprout applicable to higher ring dimensions, N.

Example 3.3 (sprout-19-20-23). *Let q_i's be 62-bit NTT primes and $r_{top} = 2^{19}\cdot r_1 \cdot r_2$, where r_1 is a 20-bit NTT prime, and r_2 is a 23-bit NTT prime. Typically, we can choose r_1 = 1,179,649 and r_2 = 8,519,681, the NTT primes for ring dimension $N \le 2^{17}$. Here, the sprouts can represent any bit length from 1 to 62 as*

$$\{2^1, \cdots, 2^{19}, r_1, 2r_1, 2^2 r_1, r_2, \cdots, 2^{19} r_2, r_2 r_1, \cdots, 2^{18} r_2 r_1\}.$$

Ciphertext Moduli with Universal Sprouts. Once a universal sprout is chosen, Algorithm 3 describes how to derive an output modulus $Q' \approx Q/\Delta$ from the input modulus Q and the rescaling factor Δ via rational rescaling. In the algorithm, the universality condition ensures that a suitable sprout r' can be found for Q'.

Algorithm 3: Selecting grafted ciphertext moduli after rational rescale.

Input: $Q = q_0 \cdots q_{\ell-1} \cdot r$ for input ciphertext with scaling factor Δ and word size $w = 2^{\omega}$.
Output: $Q' = q_0 \cdots q_{\ell'-1} \cdot r'$ for output ciphertext with $Q' \approx Q/\Delta$.
1 Choose $r' \mid r_{\text{top}}$ such that $|\log_2 r' - (\log_2(Q/\Delta) \bmod \omega)|$ achieves the minimum.
2 $\ell' \leftarrow \lfloor(\omega\cdot\ell + \lfloor\log_2 Q - \log_2\Delta\rceil)/\omega\rfloor$
return $Q' = q_0 \cdots q_{\ell'-1} \cdot r'$

Using Algorithm 3, one can always select an output modulus $Q' \approx Q/\Delta$ in Algorithm 1 as a grafted modulus dividing $Q_{\max}$, thereby enabling key switching from ciphertexts in arbitrary moduli during homomorphic computations. We also note that only a few unit moduli are eliminated during rational rescaling, and the resulting error remains roughly of the same magnitude as that of standard rescaling.

Now, we describe the overall process of the key switching operation, as well as the relinearization, in the Grafted RNS-CKKS. We decompose the largest ciphertext modulus $Q_{\max}$ into dnum number of similar-sized blocks $Q_0, \cdots, Q_{\text{dnum}-1}$, known as the gadget blocks, where $P \gtrsim \max_i Q_i$. Although the sprout can be placed in any gadget block, the gadget block containing the sprout must always participate in key switching. Therefore, we place the sprout in the bottom gadget block Q_0 so that key switching under the grafted moduli incurs no computational compromise compared to the original key switching procedure. The gadget blocks are

[6] We take 60-62 bits for primes instead of 64 bits for some efficiency reasons even in the 64-bit machines, e.g., due to key switching with dnum > 1 or lazy rescale.
[7] One may consider using a real/complex-valued Discrete Fourier Transform (DFT) instead of the NTT. However, it introduces implementation difficulties due to the need for higher precision than the native word size.
[8] Even for larger dimensions, we can use them by utilizing incomplete NTTs instead of complete NTTs.
[9] Among the 64 divisors, we select and use only 61 divisors. This is to avoid unnecessary modulus switchings, e.g., between $2r_1 \cdot 2^{14} \approx 2^{30}$ and $r_2 \approx 2^{30}$.

defined as:

$$Q_0 = r_{\text{top}} \cdot q_0 \cdots q_{\alpha-2},$$
$$Q_i = q_{\alpha i-1} q_{\alpha i} \cdots q_{\alpha(i+1)-2} \text{ for } 1 \le i \le \text{dnum} - 1,$$

where $L+1 = \text{dnum} \cdot \alpha$. Then, a grafted ciphertext modulus with sprouts can be represented as

$$\begin{aligned} Q = r \cdot q_0 q_1 \cdots q_{\ell-1} &= (r \cdot q_0 \cdots q_{\alpha-2}) \cdot (q_{\alpha-1} \cdots q_{2\alpha-2}) \\ &\quad \cdots (q_{\alpha(d-1)-1} \cdots q_{\ell-1}) \\ &= Q'_0 \cdot Q_1 \cdots Q_{d-2} \cdot Q'_{d-1}, \end{aligned}$$

for some $1 \le \ell \le L$, $Q'_0 \mid Q_0$, and $Q'_{d-1} \mid Q_{d-1}$ where $d = \lceil (\ell+1)/\alpha \rceil$. This choice clearly minimizes the number of gadgets for each ciphertext modulus Q.

Following the original RNS-CKKS key-switching procedures described in Section 2.3.4, we present its extension to grafted ciphertext moduli in Algorithm 4.

Algorithm 4: Key switching in grafted moduli.

Input: $\text{ct} = (\mathbf{b}, \mathbf{a}) \in \mathcal{R}_Q^2$ for $Q = Q'_0 \cdot Q_1 \cdots Q_{d-2} \cdot Q'_{d-1}$ with a switching key $\text{swk} = \{\text{swk}_i\} \in \mathcal{R}_{PQ_{\max}}^{2\times d}$.
Output: $\text{ct}' \in \mathcal{R}_Q^2$ with a switched secret key.

1 $\mathbf{a} \leftarrow \text{Inv-RS}_{Q_{\text{inter}}/Q}(\mathbf{a})$ ▷ Inv-RS to $Q_{\text{inter}} = Q_0 \cdots Q_{d-1}$
2 $\mathbf{a}_i \leftarrow \text{ModUp}_{Q_i \to PQ_{\text{inter}}}(\mathbf{a} \mod Q_i)$ for $i \in [d]$;
3 $\text{ct} \leftarrow \sum_i \mathbf{a}_i \cdot (\text{swk}_i \mod PQ_{\text{inter}})$;
4 $\text{ct}' \leftarrow \text{RS}_{PQ_{\text{inter}}/Q}(\text{ct})$ ▷ RS to Q
5 $\text{ct}' \leftarrow \text{ct}' + (\mathbf{b}, 0)$;
return ct'

3.3 Modulo Arithmetic with Power-of-two Sprouts

The correctness of multiplication in Algorithm 1 and key switching in Algorithm 4 under grafted ciphertext moduli follows directly from the correctness of modulus switching, including rational rescale, ModUp, and ModDown processes. However, when a power-of-two modulus in the universal sprout r_{top} is involved, a new type of issue arises during modulus switching. The original RNS-CKKS scheme does not cover such cases, as modulus switching was originally defined only between relatively prime bases.

In this section, we extend modulus switching algorithms so that grafted RNS-CKKS supports RNS factors with power-of-two factors. We begin by identifying cases where such extensions apply naturally.

If the exponent of the power-of-two component remains unchanged between the input and output moduli, then all CRT factors—including the power-of-two modulus—remain pairwise relatively prime. In this case, the fast basis conversion (FBC) and its applications—RS (by an integer factor), Inv-RS, and ModUp —can be used without modification, as in the original RNS-CKKS.

When the exponent differs, we handle the switching in two stages using an intermediate modulus Q_{inter}. First, we apply standard modulus switching from Q to Q_{inter}, ensuring that their power-of-two exponents are equal. Then, we use Inv-RS or RS from Q_{inter} to Q' to adjust the exponent. This order can also be reversed.

This two-step approach introduces negligible overhead, and often improves error control by separating exponent adjustment from standard switching.

Accordingly, we present new algorithms for Inv-RS and RS (including ModDown) to support such cases, as shown in Algorithms 5 and 6.

Algorithm 5: Inv-RS with power-of-two sprouts.

Input: $\mathbf{a} \in \mathcal{R}_Q$ for $Q = \tilde{Q} \cdot 2^\beta$ and $R = \tilde{R} \cdot 2^\gamma$, where $\tilde{Q}$ and $\tilde{R}$ are odd integers and $\beta, \gamma \in \mathbb{Z}_{\ge 0}$.
Output: $\text{Inv-RS}_R(\mathbf{a}) \in \mathcal{R}_{QR}$.

1 $\mathbf{b} \leftarrow \begin{cases} [2^\gamma]_{q_i} \cdot [\mathbf{a}]_{q_i} & (\text{mod } q_i) \text{ for } i \in [\ell], \\ 2^\gamma \cdot [\mathbf{a}]_{2^\beta} & (\text{mod } 2^{\beta+\gamma}), \end{cases}$ ▷ $\mathbf{b} \in \mathcal{R}_{\tilde{Q} \cdot 2^{\beta+\gamma}}$
2 $\mathbf{c} \leftarrow \text{Inv-RS}_{\tilde{R}}(\mathbf{b})$ ▷ $\mathbf{c} \in \mathcal{R}_{QR}$
return $\mathbf{c}$

In Algorithm 5, the intermediate polynomial satisfies $\mathbf{b} \equiv 2^\gamma \cdot \mathbf{a} \pmod{\tilde{Q} \cdot 2^{\beta+\gamma}}$. Then, the Inv-RS with the integer factor $\tilde{R}$ can be applied since $\tilde{R}$ is an odd integer, resulting in an output $\mathbf{c} \equiv \tilde{R} \cdot (2^\gamma \cdot \mathbf{a}) \equiv R \cdot \mathbf{a} \pmod{QR}$ as desired.

Algorithm 6: RS with power-of-two sprouts.

Input: $\mathbf{a} \in \mathcal{R}_{QR}$ for $Q = \tilde{Q} \cdot 2^\beta$ and $R = \tilde{R} \cdot 2^\gamma$, where $\tilde{Q}$ and $\tilde{R}$ are odd integers and $\beta, \gamma \in \mathbb{Z} \ge 0$.
Output: $\text{RS}_R(\mathbf{a}) \in \mathcal{R}_Q$.

1 $\mathbf{b} \leftarrow \text{ModDown}_{QR \to \tilde{Q} \cdot 2^{\beta+\gamma}}(\mathbf{a})$ ▷ $\mathbf{b} \in \mathcal{R}_{\tilde{Q} \cdot 2^{\beta+\gamma}}$
2 $\mathbf{c} \leftarrow \begin{cases} [2^{-\gamma}]_{q_i} \left([\mathbf{b}]_{q_i} - [\mathbf{b}]_{2^\gamma} \right) & (\text{mod } q_i), \\ \left([\mathbf{b}]_{2^{\beta+\gamma}} - [\mathbf{b}]_{2^\gamma} \right) / 2^\gamma & (\text{mod } 2^\beta), \end{cases}$ ▷ $\mathbf{c} \in \mathcal{R}_Q$
return $\mathbf{c}$

In Algorithm 6, the intermediate polynomial $\mathbf{b}$ satisfies $\|\mathbf{b} - \tilde{R}^{-1} \cdot \mathbf{a}\|_\infty \le k/2$, where k is the number of decomposed factors of $\tilde{R}$. We note that the last division for modulo 2^β can be replaced by shifting γ bits. The final output $\mathbf{c}$ satisfies $\mathbf{c} = (\mathbf{b} - [\mathbf{b}]_{2^\gamma})/2^\gamma$, leading to the bound $\|\mathbf{c} - 2^{-\gamma} \cdot \mathbf{b}\|_\infty \le 1/2$. This implies the overall error satisfies $\|\mathbf{c} - R^{-1} \cdot \mathbf{a}\|_\infty \le 1/2 + 2^{-\gamma} \cdot k/2$, as desired.

Lastly, we note that ModUp is used only within the key switching operation, where the power-of-two component is already adjusted by the preceding Inv-RS (line 1) of Algorithm 4. As a result, ModUp (line 2) can be applied without modification.

With the above extensions in place, we can now apply the key switching in Algorithm 4 and the homomorphic multiplication in Algorithm 1 directly under the universal sprout setting. We remark that the key switching procedure can be slightly modified to omit the Inv-RS by the power-of-two factors. That is, key switching in Algorithm 4 for the power-of-two modulus part can be performed modulo $2^{\text{ord}_2(Q)}$ by reducing the modulus, rather than modulo $2^{\text{ord}_2(Q_{\max})}$ with an additional Inv-RS.

4 Applications

Grafting optimally packs the modulus using mostly machine word-sized factors, resulting in fewer RNS components. This reduction

not only accelerates homomorphic operations but also decreases the ciphertext and public key sizes, roughly in proportion to the reduced number of factors. Also, when using Grafting, we do not need to suffer from the lack of small NTT primes or scale factors larger than the machine's word size, which naturally allows lower—and/or higher-precision computation. In addition, Grafting is particularly effective in applications that use smaller-than-usual scale factors where the original number of RNS factors is far from optimal.

In this section, we revisit CKKS applications with Grafting. In Section 4.1, we first suggest grafted parameters for the standard RNS-CKKS parameters for SHE and FHE, showcasing the expected speed-ups gained from Grafting. Then, in Section 4.2, we revisit the Bit-CKKS gate bootstrappings of Bae et al. [4] with Grafting, which benefit from the small-sized scale factors $\approx$ 28-30 bits. Finally, in Section 4.3, we revisit homomorphic comparison [18] using Grafting, as an example of Newton-like methods accelerated via adaptively chosen scale factors.

4.1 Grafted SHE and FHE Parameters

We revisit various CKKS parameters of SHE and FHE in the literature standard, specifically from the white paper of Bossuat et al. [8], which specifies the guidelines for securely implementing HE schemes and libraries such as HEaaN [21] and OpenFHE [2]. In Table 2, we summarize the details of the parameter sets, including SHE parameters in $\log_2 N = 14$ and 15 and FHE parameters in $\log_2 N = 15$ and 16. The security guideline white paper introduces several parameters basically based on Lattigo [33] and OpenFHE, but with full ternary secret keys, meaning that the coefficients of the secret keys are randomly sampled from $\{-1, 0, 1\}$. The security parameters are set as $\lambda \gtrsim 128$ for all the parameter sets, and a parameter set from [8] has $\lambda \approx 192$. The sizes of the RNS factors consisting of the ciphertext modulus are shown for each subprocedure as $X \times Y$, implying that it can be decomposed into Y RNS factors of size X bits each. The total number of the RNS factors (denoted as #) and the rank for gadget decompositions (denoted as d) are given in advance. We note that for simSHE15s, the larger-than-word-size moduli are used for the RNS factors requiring multiple-precision integer arithmetic [2]. We, instead, decompose them into factors smaller than word-size and use composite rescaling [1], which shows better performance in general.

In the table, we also introduce their adaptation to Grafting using the universal sprouts `sprout-15-16-30` and `sprout-19-20-23`. The maximum modulus (denoted as $\log_2 PQ$) and the modulus for multiplication (shown in parentheses for grafted parameters) in the grafted adaptation are maintained similarly to the original parameters. As an exception, the FHE parameter for $\log_2 N = 16$ has a much smaller maximum modulus than the original parameter but offers more available modulus for multiplications between the two consecutive bootstrappings. This is because the moduli for ciphertexts and for switching keys can be adjusted in a more efficient way.

Speed-up Expectations. The speed-up when using Grafting compared to the prior state-of-the-art can be expected via the ratio of the number of RNS factors. The run-time of the tensor product at level L is proportional to the number of RNS factors, $L + 1$. The run-time of the key-switching can be roughly expected as $(\mathsf{dnum} + 2)(L + K + 1)$ for K, the number of RNS factors for the temporary modulus (P), and the gadget rank dnum, when focusing on the number of NTT/iNTT operations. The speed-ups for homomorphic multiplications are expected to be similar to those of the key-switchings, as they are the dominating part of the multiplications. However, when the overall speed is very fast, e.g., at the bottom level, the ratio may differ due to slightly more expensive rescaling in Grafting. For instance, when comparing simSHE14 and graSHE14, we expect $8/7 = 1.14\times$ speed-up for tensor product and $(6 \cdot 10)/(5 \cdot 9) = 1.33\times$ speed-up for key-switchings at level $L = 7$.

For SlotToCoeff, CoeffToSlot, and EvalMod in FHE parameters, the run-time is significantly affected by the key-switching and multiplication timings. They will have a similar ratio to the key-switchings. However, as the levels are changing during the procedures, it is not easy to expect the gains. We roughly estimate the speed-up as the weighted mean of the ratios for SlotToCoeff, CoeffToSlot, and EvalMod, where each ratio is estimated as the key-switching ratio at the middle levels. For instance, when comparing simFHE15 and graFHE15, we expect $(12 \cdot 21)/(8 \cdot 14) = 2.25\times$ speed-up for key-switchings at level $L = 18$, $(9 \cdot 15)/(7 \cdot 12) = 1.61\times$ at level $L = 13$, and $(4 \cdot 6)/(4 \cdot 6) = 1\times$ at level $L = 2$. Assuming 50% of CoeffToSlot, 40% of EvalMod, and 10% of SlotToCoeff for the total timings, we can expect the bootstrapping speed-up to be 1.87×. We note that the ratio can differ in the levels, and we expect almost no gain in the bottom levels.

4.2 Bit-CKKS, Revisited

The binary bootstrapping of Bae et al. [4] assumes a ciphertext encrypts only a bit in each slot. Thus, the scale factors can be smaller than usual. Specifically, the BinBoot algorithm in [4] assumes each complex message $b + \epsilon$ for a bit $b \in \{0, 1\}$ with an error $|\epsilon| \ll 1$ and a fine-grained scale factor $q_0/2$. The EvalMod algorithm is then adapted to approximate $(b + \epsilon)/2 + I \mapsto b + o(\epsilon^2)$ via

$$f_{\mathsf{BinBoot}}(x) = \frac{1 - \cos(2\pi x)}{2},$$

so that $f_{\mathsf{BinBoot}}((b+\epsilon)/2+I) = (1-\cos(\pi b+\pi\epsilon))/2 = b+o(\epsilon^2)$ for $b \in \{0, 1\}$ and $|\epsilon| \ll 1$.[10] Similar approaches apply to boolean gates, e.g., $f_{\mathsf{nandBoot}}(x_0, x_1) = 2(1 + \cos(2\pi(x_0 + x_1) + \pi/6))/3$, sharing most of the parameters.

In Table 3, we provide the original and grafted variants of the Bit-CKKS [4] parameters utilizing `sprout-15-16-30` and `sprout-19-20-23` for ring dimensions $N = 2^{14}$ and 2^{16}, respectively. All the parameters use sparse secret encapsulation [10, 27] between a normal secret key with a Hamming weight h and a sparser secret key with a Hamming weight h', to avoid costly EvalMod and to lower the failure probability. The grafted adaptation follows the same strategy as in Table 2. As an exception, we could reduce the maximum modulus and the Hamming weight h for $N = 2^{16}$ while keeping the security parameter $\lambda \gtrsim 128$.[11]

In addition to the parameters in the prior art, we suggest another parameter set for $\log_2 N = 14$, namely graBinFHE14Opt, with optimizations for more available levels after bootstrapping: 3

[10] Since $\cos(\pi\epsilon) = 1 + o(\epsilon^2)$ and $1 + \cos(\pi + \pi\epsilon) = 2\sin^2(\pi\epsilon/2) = o(\epsilon^2)$.
[11] Estimated for grafted variants via the lattice estimator [3].

Table 2: Various HE parameters for CKKS from the security guidelines white paper [8], the HEaaN library [21], and the OpenFHE library [2], along with their grafted adaptations. All the parameters have a security parameter of $\lambda \gtrsim 128$, and specifically for simple/graSHE15, $\lambda = 192$. **The last** s **indicates using a sparse ternary secret. # and d indicate the total number of the RNS factors and the rank for gadget decompositions, respectively.**

Parameters		N	$\log PQ$	h	$\log q_i$					$\log p_i$	#	d
					Base	StC	Mult	EvalMod	CtS			
simSHE14	[8]	2^{14}	427	full	40	38×7				60×2	10	4
graSHE14					51×6					60×2	8	3
simSHE15	[8]	2^{15}	592	full	43	41×9				60×3	13	4
graSHE15					$58 + 59 \times 6$					60×3	10	3
simSHE15s	[2]	2^{15}	675	192	$105\ (52 + 53)$	$90 \times 5\ (45 \times 10)$				60×2	14	6
graSHE15s			679		$61 \times 3 + 62 \times 6$					62×2	11	5
simFHE15s	[21]	2^{15}	777	192	38	$32 + 28 \times 2$	28×5	38×8	41×3	42×2	22	10
graFHE15s			780		$38 + 61 \times 5 + 62 \times 5 + 4^*$ (137)					$61 + 62$	13	6
simFHE16	[8]	2^{16}	1734	full	45	30×3	35×10	60×12	56×4	61×5	35	6
graFHE16					$59 \times 11 + 60 \times 13$ (361)					61×5	29	5
simFHE16s	[21]	2^{16}	1555	192	58	42×3	42×9	58×9	58×3	$59 \times 3 + 60 \times 2$	30	5
graFHE16s					$36 + 58 \times 20 + 62$ (378)					$59 \times 3 + 60 \times 2$	27	5

available levels instead of 2, which was unavailable due to the lack of the NTT primes, but enabled by Grafting.

4.3 Homomorphic Comparison, Revisited

Iterative methods for finding a specific real number, such as Newton's iterative method, benefited from adaptive-precision computation in cleartext. Its encrypted counterpart can be the homomorphic computation with adaptive scale factors, changing corresponding to the iterations.

This also applies to homomorphic comparison methods [18, 34, 36], which are basically an iterative homomorphic evaluation of low-degree polynomials. The comparisons are frequently required for applications, such as machine learning, but which consume a lot of moduli, even requiring a few bootstrappings.

The homomorphic comparison function proposed in [18] is constructed of two polynomials, f and g. The composited polynomial maps $[-1, -\epsilon]$ and $[\epsilon, 1]$ to $[-1, -1 + 2^{-\alpha}]$ and $[1 - 2^{\alpha}, 1]$, respectively, which approximates the sign function in the range $[-1, 1]$. In more detail, composing g multiple times maps $[\epsilon, 1]$ to $[p, 1]$, where $p \approx 0.8$ is widely chosen. The more g is composed, the less ϵ becomes. After that, $[p, 1]$ is mapped into $[1 - 2^{-\alpha}, 1]$ by composing f multiple times. The more f is composed, the larger α becomes.

Regarding the homomorphic computation errors, indeed, composing g multiple times maps $[\epsilon, 1]$ to the interval $[p - e_g, 1]$, where e_g is the error introduced during composing g multiple times. During computing f, the error e_g can vanish. With this heuristic property, we can adjust the scaling factors per step using Grafting while maintaining the final precision. In more detail, the scale factors are chosen to ensure enough precision for each step of the homomorphic evaluation. We base this on an error analysis using the derivative of the comparison function (or its components f_3 and g_3) to estimate the required precision in plaintext. In the early stages, we use scale factors that are small but sufficient to maintain the necessary accuracy. Note that, during evaluation, the scale factors can be increased by multiplying power-of-two integers or decreased via rational rescaling.

5 Experimental Results

We implement Grafting using C++ for the non-grafted (simple) and grafted variant of RNS-CKKS. In this section, we report and compare execution times, ciphertext and key sizes, and the bootstrapping precision of our simple and grafted RNS-CKKS implementations, following the previous section's parameters.

We run our experiments on Ubuntu 24.04.1 LTS with an AMD Ryzen 7 3700X CPU and 32 GB of available memory. We disable CPU scaling, use only a single thread, and pin execution to one CPU core.

In the following, we define the *precision* of each experiment as the negative base-2 logarithm of the maximum error obtained from over 100 runs. Note that in [8], the precision is defined differently– the average of the negative base-2 logarithm of the error, i.e., $-\log_2 |\text{error}|$ averaged over the slots and executions. We will refer to this measure as *L1-precision*, primarily for comparison with the precision reported in [8]. The timings are averaged from over 100 runs.

If not specified, our experiments utilize full $N/2$ complex slots, which may be seen as N real slots.

5.1 SHE and FHE with Grafting

We display the experimental results for SHE and FHE operations in Table 4 based on the parameters in Table 2. The timings for tensoring, key-switching, homomorphic multiplications, and bootstrappings are shown in seconds, and the ratios between the simple and the grafted variants are given in parentheses. The timings are measured at the top levels for evaluations, i.e., top levels after bootstrapping for the FHE parameters. Our experiments report up to 1.58×, 1.83×, 2.01× speed-ups for homomorphic evaluations

Table 3: Parameters for Bit-CKKS [4] and their grafted variants with security parameter $\lambda \gtrsim 128$. Note that for the additional parameter graBinFHE14Opt**, the modulus consumption for each sub-procedure is given in advance.**

Parameters		N	$\log PQ$	h (h')	$\log q_i$ Base	$\log q_i$ StC	$\log q_i$ Mult	$\log q_i$ EvalMod	$\log q_i$ CtS	$\log p_i$	#	dnum
simBinFHE14	[4]	2^{14}	424	256 (32)	32	28(= 14 + 14)	26×2	32×7	29×2	33	14	13
graBinFHE14			426		$28 + 56 \times 5 + 62$ (52)					56	8	7
graBinFHE14Opt	Ours	2^{14}	424	256 (32)	26	24(= 12 + 12)	24×3	28×7	25×2	56	8	7
					$26 + 56 \times 5 + 62$ (72)							
simBinFHE16	[4]	2^{16}	1598	256 (32)	32	32(= 16 + 16)	30×28	32×7	32×2	58×7	46	3
graBinFHE16			1522	192 (32)	$34 + 62 \times 19$ (840)					62×5	25	4

for available levels after bootstrapping, and up to 1.92× speed-up for bootstrapping. Note that for sim/graFHE16s, the speed-up is low as expected–due to the larger scale factors. The experiments reported precision at least as high as or similar to the non-grafted cases, for instance, multiplication precision of 22.8 bits (L1-precision of 26.2 bits) for graSHE14, and bootstrapping precision for graFHE16 of 13.2 bits. Note that L1-precision is similar to or larger than in [8].

Table 4: Execution timings for SHE and FHE operations, given in milliseconds. The bootstrapping (BTS**) timings are given in seconds. Note that** SHE **and** FHE **parameters are abbreviated as** S **and** F**, respectively. The speed-ups are given in parentheses.**

Params	Tensor	KeySwitch	Mult	BTS
simS14	1.94	32.22	41.61	-
graS14	1.51 (1.3×)	24.16 (1.3×)	36.52 (1.1×)	-
simS15	4.55	93.39	118.86	-
graS15	3.53 (1.3×)	65.61 (1.4×)	91.95 (1.3×)	-
simS15s	5.56	127.38	171.64	-
graS15s	4.74 (1.2×)	93.42 (1.4×)	128.44 (1.3×)	-
simF15s	4.2	69.16	102.2	14.5
graF15s	2.78 (1.5×)	40.29 (1.7×)	57.28 (1.8×)	7.6 (1.9×)
simF16	12.96	248.47	360.84	86.5
graF16	8.21 (1.6×)	136.07 (1.8×)	179.87 (2.0×)	71.7 (1.2×)
simF16s	12.46	227.68	329.38	37
graF16s	9.88 (1.3×)	186.81 (1.2×)	247.45 (1.3×)	35.5 (1.1×)

In addition to the performance gain, we also present the sizes of the public keys for the basic RNS-CKKS case, following BitPacker [39], and Grafting. We estimate the relinearization key sizes in Table 2, which will allow multiplication at each level, in CRT format, since it is the standard choice for better performance. We note that for simple and grafted, the keys are naturally packed in the machine's word size, and the size of each key is already near optimal.[12] However, in BitPacker, the number of required keys is different. In general, each level requires keys in the different moduli, which should be generated individually for each level.

In Table 5, we compare each parameter set's relinearization key sizes to allow multiplications at each level. The simple and grafted cases are assumed to generate only one key at modulus $PQ_{\max}$ for whole levels, which can be reused. They differ by the ratio between the number of primes consisting of the modulus $PQ_{\max}$. In the case of BitPacker [39], we assume one key for each level (except for the SlotToCoeff and CoeffToSlot levels), but the gadget decompositions are optimized for better sizes. Note that the key sizes are computed based on the levels of the corresponding simple SHE/FHE parameters. Thus, if one wants to use a ciphertext modulus other than the multiplicative levels in the simple parameter, the key sizes will increase for BitPacker but not for Grafting.

Table 5: Size estimations for the relinearization key in SHE and FHE parameters, in MB. The changes in the key sizes in percentiles for each parameter over the simple parameters are given in parentheses.

Parameters	simple	BitPacker [39]	grafted
SHE14	10.49	29.88 (185%↑)	7.08 (32%↓)
SHE15	27.26	88.60 (225%↑)	17.30 (37%↓)
SHE15s	44.04	97.00 (122%↑)	31.46 (29%↓)
FHE15	115.34	313.00 (171%↑)	44.04 (62%↓)
FHE16	220.20	1394.61 (533%↑)	157.29 (29%↓)
FHE16s	157.29	981.47 (524%↑)	146.80 (7%↓)

5.2 Bit-CKKS with Grafting

We display the experimental results for Bit-CKKS parameters in Table 6, based on the parameters designed in Table 3. The message space for the input messages is binary, $\{0, 1\}$, and the messages are randomly chosen. We use real slots for $\log_2 N = 14$ parameters (i.e., 2^{13} binary messages) and full complex slots for $\log_2 N = 16$ parameters (i.e., 2^{16} binary messages), as in [4].

Our experiments report speed-ups of 1.81-1.89×, along with moderate precision of 6.6 and 6.1 bits for sim/graBinFHE14 and sim/graBinFHE16, respectively. Note that for the additional parameter set with optimized available levels, 3.5 bits of precision are reported. The precision is similar for both non-grafted and grafted variants. We additionally report the multiplication timings with speed-ups of up to 2.07×. For $\log_2 N = 14$, the multiplication timing measurement is performed at the near-bottom modulus, yielding

[12] For transmission, half of the key's polynomials (say, a-parts) can be replaced by the seed when stored and transformed on the fly via an extendable output function (XOF).

Table 6: Execution times for Bit-CKKS binary NAND gate bootstrapping (in seconds), multiplication (in milliseconds), and size estimations for the relinearization key (rlk, in MB).

Parameters	nandBoot	Mult	rlk size
simBinFHE14	5.18	16.1	47.71
graBinFHE14	2.74 (1.9×)	16.2 (1.0×)	16.52 (65%↓)
graBinFHE14Opt	2.75 (1.9×)	15.7 (1.0×)	16.52 (65%↓)
simBinFHE16	102.1	884.8	144.70
graBinFHE16	56.41 (1.8×)	428.1 (2.1×)	109.05 (25%↓)

almost the same timings. The relinearization key sizes are given in advance.

5.3 Homomorphic Comparison with Grafting

We display the experimental results for the homomorphic comparison in Tables 7 and 8, based on the parameters from Table 2.

We provide implementation results that primarily reduce the modulus consumption and execution timings of various homomorphic comparison functions proposed in [18] while maintaining precision. Specifically, we used seven-degree polynomials f_3 and g_3 from [18] to construct homomorphic comparison functions.

We use two different homomorphic comparison functions, comp1 and comp2, both of which compose the f and g functions, but with different numbers targeting different input ranges and precision. The input range of the homomorphic comparison function can be denoted by $[-1, -\epsilon] \cup [\epsilon, 1]$ where $\epsilon = 2^{-8}$ for comp1 and $\epsilon = 2^{-16}$ for comp2. The function f_3 is composed twice for both comparison functions. The function g_3 is composed four times for comp1 and eight times for comp2.

In the following tables, the amounts of modulus consumed per step are denoted as $X \times Y$, where X represents the number of bits consumed when computing a single polynomial (either f_3 or g_3) and Y is the number of such polynomials. Note that computing a seven-degree polynomial requires three multiplicative depths, so the scale factor used during computing a polynomial is $X/3$ bits.

Table 7 reports 10% to 27% reduction in modulus consumption compared to the simple(non-grafted) case. In practice, computing homomorphic comparison requires a few bootstrappings. Therefore, reducing modulus consumption can be directly converted into a significant speed-up by reducing the number of bootstrappings. Additionally, adjusting the available modulus for each bootstrapping using the flexible output modulus bootstrapping, the bootstrapping with lower Q_{top}, can further enhance the speed-up of each bootstrapping. Table 8 reports 1.24× to 1.56× speed up compared to the simple (non-grafted) case.

Table 7: Modulus consumption for homomorphic comparison. The per-step modulus consumption is denoted by $X \times Y$, where X represents the modulus consumed for computing a single polynomial (either f_3 or g_3) and Y represents the number of such polynomials.

Configs	Parameters	Modulus Consumption		Prec
		Total	Per-Step	
comp1	simFHE16	630	105×6	16.3
	graFHE16	531 (16%↓)	$84 \times 4 + 90 + 105$	16.5
comp2	simFHE16	1050	105×10	16.4
	graFHE16	942 (10%↓)	$93 \times 9 + 105$	16.4
comp1	simFHE16s	756	126×6	23.1
	graFHE16s	552 (27%↓)	$84 \times 4 + 90 + 126$	23.1
comp2	simFHE16s	1260	126×10	23.5
	graFHE16s	963 (24%↓)	$93 \times 9 + 126$	23.3

Table 8: Execution timings for homomorphic comparison functions with bootstrapping.

Configs	Parameters	Bootstrapping		Time(s)
		#BTS	Avail. Modulus	
comp1	simFHE16s	1	378× 1	42.7
	graFHE16s	1	216	34.3 (1.24)
comp2	simFHE16s	3	378×3	114.5
	graFHE16s	2	$372 + 219$	73.5 (1.56)

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References

[1] Rashmi Agrawal, Jung Ho Ahn, Flavio Bergamaschi, Ro Cammarota, Jung Hee Cheon, Fillipe D. M. de Souza, Huijing Gong, Minsik Kang, Duhyeong Kim, Jongmin Kim, Hubert de Lassus, Jai Hyun Park, Michael Steiner, and Wen Wang. 2023. High-precision RNS-CKKS on fixed but smaller word-size architectures: theory and application. In *Proceedings of the 11th Workshop on Encrypted Computing & Applied Homomorphic Cryptography* (Copenhagen, Denmark) *(WAHC '23)*. Association for Computing Machinery, New York, NY, USA, 23–34. doi:10.1145/3605759.3625257

[2] Ahmad Al Badawi, Jack Bates, Flavio Bergamaschi, David Bruce Cousins, Saroja Erabelli, Nicholas Genise, Shai Halevi, Hamish Hunt, Andrey Kim, Yongwoo Lee, Zeyu Liu, Daniele Micciancio, Ian Quah, Yuriy Polyakov, Saraswathy R.V., Kurt Rohloff, Jonathan Saylor, Dmitriy Suponitsky, Matthew Triplett, Vinod Vaikuntanathan, and Vincent Zucca. 2022. OpenFHE: Open-Source Fully Homomorphic Encryption Library. In *Proceedings of the 10th Workshop on Encrypted Computing & Applied Homomorphic Cryptography* (Los Angeles, CA,

USA) *(WAHC'22)*. Association for Computing Machinery, New York, NY, USA, 53–63. doi:10.1145/3560827.3563379

[3] Martin R. Albrecht, Rachel Player, and Sam Scott. 2015. On the concrete hardness of Learning with Errors. *Journal of Mathematical Cryptology* 9, 3 (2015), 169–203. doi:doi:10.1515/jmc-2015-0016

[4] Youngjin Bae, Jung Hee Cheon, Jaehyung Kim, and Damien Stehlé. 2024. Bootstrapping Bits with CKKS. In *EUROCRYPT 2024, Part II (LNCS, Vol. 14652)*, Marc Joye and Gregor Leander (Eds.). Springer, Cham, 94–123. doi:10.1007/978-3-031-58723-8_4

[5] Youngjin Bae, Jaehyung Kim, Damien Stehlé, and Elias Suvanto. 2024. Bootstrapping Small Integers With CKKS. In *ASIACRYPT 2024, Part I (LNCS, Vol. 15484)*, Kai-Min Chung and Yu Sasaki (Eds.). Springer, Singapore, 330–360. doi:10.1007/978-981-96-0875-1_11

[6] Jean-Claude Bajard, Julien Eynard, M. Anwar Hasan, and Vincent Zucca. 2016. A Full RNS Variant of FV Like Somewhat Homomorphic Encryption Schemes. In *SAC 2016 (LNCS, Vol. 10532)*, Roberto Avanzi and Howard M. Heys (Eds.). Springer, Cham, 423–442. doi:10.1007/978-3-319-69453-5_23

[7] Mariya Georgieva Belorgey, Sergiu Carpov, Nicolas Gama, Sandra Guasch, and Dimitar Jetchev. 2023. Revisiting Key Decomposition Techniques for FHE: Simpler, Faster and More Generic. Cryptology ePrint Archive, Paper 2023/771. https://eprint.iacr.org/2023/771

[8] Jean-Philippe Bossuat, Rosario Cammarota, Ilaria Chillotti, Benjamin R. Curtis, Wei Dai, Huijing Gong, Erin Hales, Duhyeong Kim, Bryan Kumara, Changmin Lee, Xianhui Lu, Carsten Maple, Alberto Pedrouzo-Ulloa, Rachel Player, Yuriy Polyakov, Luis Antonio Ruiz Lopez, Yongsoo Song, and Donggeon Yhee. 2024. Security Guidelines for Implementing Homomorphic Encryption. Cryptology ePrint Archive, Paper 2024/463. https://eprint.iacr.org/2024/463

[9] Jean-Philippe Bossuat, Christian Mouchet, Juan Ramón Troncoso-Pastoriza, and Jean-Pierre Hubaux. 2021. Efficient Bootstrapping for Approximate Homomorphic Encryption with Non-sparse Keys. In *EUROCRYPT 2021, Part I (LNCS, Vol. 12696)*, Anne Canteaut and François-Xavier Standaert (Eds.). Springer, Cham, 587–617. doi:10.1007/978-3-030-77870-5_21

[10] Jean-Philippe Bossuat, Juan Ramón Troncoso-Pastoriza, and Jean-Pierre Hubaux. 2022. Bootstrapping for Approximate Homomorphic Encryption with Negligible Failure-Probability by Using Sparse-Secret Encapsulation. In *ACNS 22International Conference on Applied Cryptography and Network Security (LNCS, Vol. 13269)*, Giuseppe Ateniese and Daniele Venturi (Eds.). Springer, Cham, 521–541. doi:10.1007/978-3-031-09234-3_26

[11] Zvika Brakerski. 2012. Fully Homomorphic Encryption without Modulus Switching from Classical GapSVP. In *CRYPTO 2012 (LNCS, Vol. 7417)*, Reihaneh Safavi-Naini and Ran Canetti (Eds.). Springer, Berlin, Heidelberg, 868–886. doi:10.1007/978-3-642-32009-5_50

[12] Zvika Brakerski, Craig Gentry, and Vinod Vaikuntanathan. 2012. (Leveled) fully homomorphic encryption without bootstrapping. In *Proceedings of the 3rd Innovations in Theoretical Computer Science Conference (ITCS '12)*. Association for Computing Machinery, New York, NY, USA, 309–325. doi:10.1145/2090236.2090262

[13] Jung Hee Cheon, Wonhee Cho, Jaehyung Kim, and Damien Stehlé. 2023. Homomorphic Multiple Precision Multiplication for CKKS and Reduced Modulus Consumption. In *ACM CCS 2023*, Weizhi Meng, Christian Damsgaard Jensen, Cas Cremers, and Engin Kirda (Eds.). ACM Press, 696–710. doi:10.1145/3576915.3623086

[14] J. H. Cheon, H. Choe, M. Kang, and J. Kim. 2024. Grafting: Complementing RNS in CKKS. Cryptology ePrint Archive, Paper 2024/1014.

[15] Jung Hee Cheon, Hyeongmin Choe, and Jai Hyun Park. 2025. Tree-based lookup table on batched encrypted queries using homomorphic encryption. *Journal of the Korean Mathematical Society* 62, 5 (Sept. 2025), 1237–1263. doi:10.4134/JKMS.j240423

[16] Jung Hee Cheon, Kyoohyung Han, Andrey Kim, Miran Kim, and Yongsoo Song. 2019. A Full RNS Variant of Approximate Homomorphic Encryption. In *SAC 2018 (LNCS, Vol. 11349)*, Carlos Cid and Michael J.: Jacobson, Jr. (Eds.). Springer, Cham, 347–368. doi:10.1007/978-3-030-10970-7_16

[17] Jung Hee Cheon, Andrey Kim, Miran Kim, and Yong Soo Song. 2017. Homomorphic Encryption for Arithmetic of Approximate Numbers. In *ASIACRYPT 2017, Part I (LNCS, Vol. 10624)*, Tsuyoshi Takagi and Thomas Peyrin (Eds.). Springer, Cham, 409–437. doi:10.1007/978-3-319-70694-8_15

[18] Jung Hee Cheon, Dongwoo Kim, and Duhyeong Kim. 2020. Efficient Homomorphic Comparison Methods with Optimal Complexity. In *ASIACRYPT 2020, Part II (LNCS, Vol. 12492)*, Shiho Moriai and Huaxiong Wang (Eds.). Springer, Cham, 221–256. doi:10.1007/978-3-030-64834-3_8

[19] Hyeongmin Choe. 2025. *Accelerating Homomorphic Computation through Machine-Efficient Arithmetic*. Ph.D. Dissertation. Seoul National University, Seoul, South Korea. https://snu-primo.hosted.exlibrisgroup.com/permalink/f/1fgjt92/82SNU_INST51966945280002591 Electronic resource.

[20] Chi-Ming Marvin Chung, Vincent Hwang, Matthias J. Kannwischer, Gregor Seiler, Cheng-Jhih Shih, and Bo-Yin Yang. 2021. NTT Multiplication for NTT-unfriendly Rings: New Speed Records for Saber and NTRU on Cortex-M4 and AVX2. *IACR Transactions on Cryptographic Hardware and Embedded Systems* 2021, 2 (Feb. 2021), 159–188. doi:10.46586/tches.v2021.i2.159-188

[21] CryptoLab. 2022. HEaaN Library. Available at https://heaan.it/.

[22] Nir Drucker, Guy Moshkowich, Tomer Pelleg, and Hayim Shaul. 2024. BLEACH: Cleaning Errors in Discrete Computations Over CKKS. *Journal of Cryptology* 37, 1 (Jan. 2024), 3. doi:10.1007/s00145-023-09483-1

[23] Junfeng Fan and Frederik Vercauteren. 2012. Somewhat Practical Fully Homomorphic Encryption. Cryptology ePrint Archive, Report 2012/144. https://eprint.iacr.org/2012/144

[24] Craig Gentry, Shai Halevi, and Nigel P. Smart. 2012. Homomorphic Evaluation of the AES Circuit. In *CRYPTO 2012 (LNCS, Vol. 7417)*, Reihaneh Safavi-Naini and Ran Canetti (Eds.). Springer, Berlin, Heidelberg, 850–867. doi:10.1007/978-3-642-32009-5_49

[25] Shai Halevi, Hamish Hunt, Victor Shoup, Oliver Masters, Flavio Bergamaschi, Jack Crawford, Fabian Boemer, et al. 2021. HElib (version 2.2.1). Available at https://github.com/homenc/HElib.

[26] Shai Halevi and Victor Shoup. 2020. Design and implementation of HElib: a homomorphic encryption library. Cryptology ePrint Archive, Paper 2020/1481. https://eprint.iacr.org/2020/1481

[27] Shai Halevi and Victor Shoup. 2021. Bootstrapping for HElib. *Journal of Cryptology* 34, 1 (Jan. 2021), 44 pages. doi:10.1007/s00145-020-09368-7

[28] Kyoohyung Han and Dohyeong Ki. 2020. Better Bootstrapping for Approximate Homomorphic Encryption. In *CT-RSA 2020 (LNCS, Vol. 12006)*, Stanislaw Jarecki (Ed.). Springer, Cham, 364–390. doi:10.1007/978-3-030-40186-3_16

[29] Andrey Kim, Antonis Papadimitriou, and Yuriy Polyakov. 2022. Approximate Homomorphic Encryption with Reduced Approximation Error. In *CT-RSA 2022 (LNCS, Vol. 13161)*, Steven D. Galbraith (Ed.). Springer, Cham, 120–144. doi:10.1007/978-3-030-95312-6_6

[30] Jaehyung Kim, Jinyeong Seo, and Yongsoo Song. 2024. Simpler and Faster BFV Bootstrapping for Arbitrary Plaintext Modulus from CKKS. In *Proceedings of the 2024 ACM SIGSAC Conference on Computer and Communications Security (CCS '24)*. Association for Computing Machinery, New York, NY, USA, 2535–2546. doi:10.1145/3658644.3670302

[31] Miran Kim, Dongwon Lee, Jinyeong Seo, and Yongsoo Song. 2023. Accelerating HE Operations from Key Decomposition Technique. In *CRYPTO 2023, Part IV (LNCS, Vol. 14084)*, Helena Handschuh and Anna Lysyanskaya (Eds.). Springer, Cham, 70–92. doi:10.1007/978-3-031-38551-3_3

[32] Seonghak Kim, Minji Park, Jaehyung Kim, Taekyung Kim, and Chohong Min. 2022. EvalRound Algorithm in CKKS Bootstrapping. In *Advances in Cryptology – ASIACRYPT 2022*. Springer Nature Switzerland, Cham, 161–187.

[33] Lattigo 2024. Lattigo v6. Online: https://github.com/tuneinsight/lattigo. EPFL-LDS, Tune Insight SA.

[34] Eunsang Lee, Joon-Woo Lee, Young-Sik Kim, and Jong-Seon No. 2022. Optimization of Homomorphic Comparison Algorithm on RNS-CKKS Scheme. *IEEE Access* 10 (2022), 26163–26176. doi:10.1109/ACCESS.2022.3155882

[35] Eunsang Lee, Joon-Woo Lee, Junghyun Lee, Young-Sik Kim, Yongjune Kim, Jong-Seon No, and Woosuk Choi. 2022. Low-Complexity Deep Convolutional Neural Networks on Fully Homomorphic Encryption Using Multiplexed Parallel Convolutions. In *Proceedings of the 39th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 162)*. PMLR, 12403–12422. https://proceedings.mlr.press/v162/lee22e.html

[36] Eunsang Lee, Joon-Woo Lee, Jong-Seon No, and Young-Sik Kim. 2022. Minimax Approximation of Sign Function by Composite Polynomial for Homomorphic Comparison. *IEEE Transactions on Dependable and Secure Computing* 19, 6 (2022), 3711–3727. doi:10.1109/TDSC.2021.3105111

[37] Johannes Mono and Tim Güneysu. 2023. A New Perspective on Key Switching for BGV-like Schemes. Cryptology ePrint Archive, Paper 2023/1642. https://eprint.iacr.org/2023/1642

[38] Donghwan Rho, Taeseong Kim, Minje Park, Jung Woo Kim, Hyunsik Chae, Ernest Ryu, and Jung Hee Cheon. 2025. Encryption-Friendly LLM Architecture. In *International Conference on Representation Learning*, Y. Yue, A. Garg, N. Peng, F. Sha, and R. Yu (Eds.), Vol. 2025. 41638–41664. https://proceedings.iclr.cc/paper_files/paper/2025/file/6715b4e97be055687c1ecaf33913d358-Paper-Conference.pdf

[39] Nikola Samardzic and Daniel Sanchez. 2024. BitPacker: Enabling High Arithmetic Efficiency in Fully Homomorphic Encryption Accelerators. In *Proceedings of the 29th ACM International Conference on Architectural Support for Programming Languages and Operating Systems, Volume 2* (La Jolla, CA, USA) *(ASPLOS '24)*. Association for Computing Machinery, New York, NY, USA, 137–150. doi:10.1145/3620665.3640397

[40] SEAL 2023. Microsoft SEAL (release 4.1). https://github.com/Microsoft/SEAL. Microsoft Research, Redmond, WA.

[41] Hyewon Sung, Sieun Seo, Taekyung Kim, and Chohong Min. 2024. EvalRound+ Bootstrapping and its Rigorous Analysis for CKKS Scheme. Cryptology ePrint Archive, Paper 2024/1379. https://eprint.iacr.org/2024/1379